

3rd Applied Active Inference Symposium ~ "Enacting Ecosystems of Shared Intelligence" ~ 1st session

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hello and welcome everyone this is the third applied octave inference Symposium enacting ecosystems of shared intelligence it is August 22nd 2023 we have a lot of really awesome presentations and sessions coming up in the first interval and in the second interval so let's just get right into it with our first talk this will be by Andre bastos so Andre thank you for joining and looking forward to your talk thank you very well Daniel and thank you for the other organizers of the active inference uh conference and Symposium here I'm glad to be talking to you guys today about my work it's the title of the presentation is multi-laminar multi-area recordings under non-human primate that's these guys right here in CAC monkeys suggests that predictive coding is implemented by a predictive routing and so what do we mean by these things well uh just as a point of introduction I'd like to start by just saying that um we we started out uh I started out working with with Carl Tristan in about 2010 and we were really interested in trying to map the neurobiology of the predictive coding circuits that have that have been theorized and hypothesized onto cortical laminar architecture which you can see here and uh in in that work with him we proposed that different layers and different oscillations were involved in either capturing and calculating prediction error or predictions and so what what we do in my lab and I recently started as an assistant professor in the department of psychology at Vanderbilt and so what we do is we train rhesus macaque monkeys on on tasks which are more or less predictable and that have have different sensory elements that are more or less predictable and we'll get into what that looks like and then we use multi-laminar probes so high density probes with dozens or hundreds of contacts that can record across different layers and so this is a beautiful drawing here of what what the layers look like of Cortex by Santiago ramon y Cajal over 100 years ago and you can see this really beautiful layer layered architecture layer one two three four is this dense one in the middle five you get these large pyramidal neurons and then six on the bottom and so we're interested in in

my lab and primarily at two levels of explanation the first is what do these neurons Spike to what makes them fire Action potentials and excites them and do they get more or less excitable depending on how predictable stimuli are and second how do they oscillate and which frequency bands do those rhythmic activity tend to cause them to fire more or less action potentials and so we record across layers in one area over here but we might put in our electrodes and several other areas so that we can also capture aspects of cortical communication and these oscillations in this communication we think is is really intimately linked to the interaction between inhibitory cells these cells in red here and excitatory neurons these these cells in Black that are doing okay so without further ado let's just dive right into it here so here's the outline of my talk in part one we're going to be talking about first an overview of what do we actually mean by this predictive coding model and how do we think it it may be mapped onto cortex and paying special attention to the neurophysiology so here's the bird's eye overview of predictive coding at the level that maybe a neurobiologist or a cognitive scientist or a um or a cognitive neuroscientist might need to know about in order to be able to uh to apply it to to their own work and so here's uh here's what I mean by predictive coding that there's a hypothesized circuit that is especially rich in higher order areas so here's like the front of the brain which learns about the statistical regularities of the world and creates predictions about what is going to be seen and felt and touched and so on in the next moment in time and in this moment in time and sends those predictions to earlier parts of the brain so it's a lower order cortex receive these predictions and sensory inputs the inputs coming from the eyes and ears and all of our sensorium has been compared to this top-down prediction and if there's a mismatch that's a prediction error and the prediction error is fed forward it's in this match response and it can be used to drive models Drive these internal representations in order to be better over time and to make better predictions and in particular the these prediction errors are hypothesized to be housed in superficial layers of Cortex and so that's a that's a hypothesis that we've been able to test now with these laminar electrodes so let's see how our experimental design worked in order for us to be able to look at the difference between predictable and unpredictable sensory stimuli so we train monkeys on what's called a delayed match to sample tasks it's a classic task of working memory where monkeys are trained to fixate on the center of the screen then there's a visual stimulus in this case a car stimulus that's shown there's a brief delay in which that stimulus has to be held in working memory when the animal is tested on which of the three items was in working memory the animal makes a correct response when they cica in this case to the upper left to indicate that the car stimulus

was last shown so we then manipulated the predictability of this information that was to enter working memory in the following way for 50 trials in a row we would either keep the sample stimulus uh constant so car car and that's being depicted by the a here or in another context there would be an unpredictable sampling regime where either the car orange or green could be sampled at any moment in time so the animals went back and forth in being in a predictable context or an unpredictable context and then we asked the question what does grain activity look like when the exact same stimulus is processed in a predictable regime versus an unpredictable regime and in order to do this work we implanted these monkeys with multi-laminar probe so these are probes again with these at that time uh that I did this work about a dozen or so or maybe up to 32 contacts nowadays we can go into the several hundreds of contacts along a single probe we can use MRI guidance in order to position those probes perpendicular to Cortex such that they'll span each of the six cortical layers that are one through six here you can see the white matter here below and then we sampled from different parts of both the visual cortex because remember those are the areas that might issue prediction errors in the face of an unpredictable sample as well as higher order cortex which has been hypothesized to create these internal models that have been issue predictions so here's what we found we first just analyzed the multi-unit activity which is a measure of the spike rate so how many Action potentials per unit of time are these neurons firing and what's being shown here is the multi-unit activity as a difference between when that sample was unpredictable minus predictable so positive activity means more activity during an unpredictable sample I.E a violation of a prediction or a putative prediction error signal and all throughout the brain from V4 the sensory area all the way to these higher order frontal parietal regions we saw without any exception that neurons were much more excitable when they were processing an unpredictable stimulus compared to predictable stimulus and also not surprisingly this started already at the feed forward sweep of activity such that these red bars here indicate the start of significance so sure enough area V4 was the first to show significance so more spiking to the unpredictable overpredictable and this activity fed forward up the cortical hierarchy next we looked not only at the firing rate of these neurons but also the local field potential which is informative about the local oscillatory activity that's present so you can think of these oscillations as massive waves of activity that are synchronous across the large population of neurons but of course are produced through the interplay between these recurrent inhibitory bouts of activity and so here's what we found We performed this power spectral analysis again comparing power in the unpredictable case versus Power in the

predictable case and we did it we did a percent change metric very simple percent change metric so again what's shown here on the on the y-axis is in positive more power to unpredictable samples and in the negative values more power to the predictable samples now what we saw in the gamma range so that's everywhere above about 30 or 40 Hertz again in all areas very strongly mirrored what we saw in the spiking activity namely but there was more of this high frequency and also the classic low range frequency in the gamma range so in both the low gamma and high gamma ranges an enhancement of power during the unpredictable cues and that was also the case in the Theta band and this makes somewhat sense from the point of view of neurophysiology that this Theta frequency and this gamma frequency they very often track one another and do similar things in cortex why exactly that is the case we actually don't know uh uh what the what the reason for that is but that's an observation that we've seen in many other tasks but what we saw in between the Theta and the gamma bands was this other band at between 10 to about 20 or slightly higher in frequency this classic Alpha Beta band and this Alpha Beta band this Alpha Beta band uh acted in the exact opposite way namely there was more power to predictable samples closed captioning not available okay so next we had these laminar electrodes so let's look at the the difference between what these neurons are doing when you record from the so-called superficial layers of Cortex versus the so-called deep layers of Cortex so there so the superficial layers are layers two three the Deep layers are these layers five and six uh these superficial layers they are the layers that feed forward so remember from that introductory slide that these are the layers that were hypothesized based on predictive coding Theory to house the prediction error neurons because according to Theory those prediction error signals are sent in the feed forward Direction so we we then tested that in a second way which was to test whether or not those enhanced signals during unpredictable samples are more present in layer 2 3 and that's exactly what we found in V4 so when we again take the spiking signal and subtract unpredictable minus predictable spiking we see that this the signal is more enhanced in superficial layers of V4 there was a trend towards that in PFC as well but in V4 it was a significant observation and furthermore when we now look at the lfp gamma power this power in the 40 to 90 Hertz band again we do the the power change so these positive signals mean there's an enhancement of gamma power to the unpredictable signal and again now in both V4 and PFC it's The Superficial layers that are having more of this unpredictable signal again and that's consistent with the theorized type of circuitry for for the for these canonical circuits uh next we I we we were interested in whether or not these power changes were blankets across the board that just overall increased uh new

activity and Gamma activity independent of the exact sample that was being shown or whether these signals were actually specific to the actual item that was being predicted because remember in our task design animals were for a large portion of time predicting that there should be a car stimulus or predicting an orange or predicting the green stimulus so in that particular task requirement it wouldn't make sense if um if task representations for all three stimuli all went up to an unpredictable stimuli that would be more consistent with just an overall arousal signal but what we were next interested in is whether or not neurons that actually were selected for the car if it was those neurons in particular that had the unpredictable signals and what we found is that that was indeed the case so we performed the same analysis as before lfp gamma power difference for unpredictable minus predictable but now as a function of whether or not those neurons preferred the stimulus that was being predicted and what we found is that the preference actually modulated this gamma band activity very strongly and so it was this this unpredictable signal change was more specific to neurons that actually preferred that predicted stimulus indicating that this prediction was not acting as a blanket similar to an arousal mechanism but was actually reaching down at the level of the representations that were specific to the thing that needed to be predicted and that that was even more the case in the beta frequency range actually where the neurons that didn't encode the predicted stimulus they couldn't care less whether or not that stimulus was uh was predictable it was really only neurons that that carried information about the stimulus that was being predicted those were the neurons that were were not enhancing but diminishing their beta during an unpredictable signal or set another way enhancing their beta to a predicted signal change so here's what we think is going on we call we call this neurophysiological implementation that we found of predictive coding we call it predictive routing to make a semantic point but also a slightly more neurophysiological and computational argument that we don't actually find strong evidence for this mismatch and prediction signal at the level of neurons but instead we think that this prediction um circuit exists but uses these oscillations in order to implement what is predicted so here's how we think this works so when a prediction matches a stimulus as is being shown here when you're predicting a and you get a stimulus a then the a column the in other words it doesn't have to be a column but the the area of the brain or the neurons that encode a are going to become prepared to receive the stimulus a by enhancing the power of this Alpha Beta signal so this top-down data from higher order areas we think prepares a cortical circuit to receive a predicted stimulus and It prepares that stimulus by enhancing that's this up Arrow here the alpha beta especially in the Deep layers which can

inhibit both the gamma and the spiking signal in The Superficial layers as a result um so this top-down signal inhibits The Superficial layers and as a result there is decreased gamma and spiking and decreased feed forward output in the case of a predictable stimulus the case of an unpredictable stimulus well stimulus a just like before is still going to be coming in but because it's not predictable that column that set of neurons that normally represents a was not previously prepared by an alpha beta signal and so the alpha beta signal is relatively low in an unpredictable context releasing the inhibition that the Deep layers exert on superficial layers and as a result causing more gamma more spiking and more feed forward output which in other words what we are hypothesizing here is that predictive coding and this error computation comes about as a failure in predictive inhibition and as a result of the lack of these mechanisms being in place during a unpredictable stimulus there is an enhanced gamma output and that's what we call a prediction error that's what we interpret as a prediction error so here's the predictive routing model one more time in a predictable stimulus top-down inputs prepare the column to receive a stimulus that preparation leads to inhibition less feed forward output in a unpredictable case those mechanisms aren't in place as a result more of the forward gamma output so that's part one that's our cortical neurophysiological hypothesis then that we call predictive routing of how we believe predictive coding is implemented in the brain and next I want to turn to a little bit more about how these rhythms are actually produced in the brain because one of the things that we noticed from just placing our laminar electrodes into cortex is that independent of where we put our probe so here's an example of prefrontal Cortex activity we would see these high frequency gamma waves and superficial layers and slower frequency Alpha Beta waves in these deep layers and you can you can do this analysis very easily and quickly to reveal this what we call spectrol laminar pattern and what this is is simply dividing each contact on the probe dividing the power that you observe at that contact by the maximum by the contact with the maximum power along the probe and what this does is it normalizes power such that a value of one is the layer that had the most power at that particular frequency and so if you look here on the x-axis this is frequency so these red colors are tracking which particular layer has more power at that frequency so you see more of this low and high gamma power in The Superficial layer so that's above point zero on the y-axis and more of this Alpha Beta activity in the deeper layers of the cortex and so in a um in a large scale um in a large scale a survey of Cortex uh together with Bob desimone's lab and uh Diego and so I did this partially while I was at MIT when I was a postdoc with Earl Miller and I continued this now and at Vanderbilt so this this was a really cross-institutional study and

involved multiple labs and the reason that that was important was that most Labs specialize in being a prefrontal cortex lab or a V4 lab or a lip lab Etc and relatively few investigators had actually done the necessary work to be able to record with these laminar probes in multiple areas simultaneously and so therefore by coordinating and pooling data from multiple experiments we were able to record from a total of 14 areas in the study that we have now now in review and what we found was that independent of the area we found the spectral laminar pattern that was highly concerned served across areas and so here it is from 0 to 150 Hertz for area V4 and the sensory cortex 7A in parietal cortex prefrontal cortex in the front of the brain everywhere we looked we found that there was this enhanced gamma power in these red lines here in superficial layers and enhanced Alpha Beta power into deeper layers and what we what we then turn to next is first of all well let's really confirm um at a histological anatomical level that it's really that that is really these layers that it's is it layer one is it Layer Two is it four that expresses gamma um and the same question for beta let's really confirm that that is in fact the case because it seemed like a very prominent um hypothesis and a very prominent observation we wanted to to Really confirm it um so that's the first part of this uh part two and then we're going to get into um which specific mechanism might generate these oscillations and how does that inform inform computational modeling and so then this this was work done in my lab at Vanderbilt led by a very talented undergraduate student actually Max lichtenfeld and so we performed these histological reconstructions here's the probe an example probe where we perform these tiny electrolytic lesions in particular areas while we were recording here's the overlay of the laminar boundaries so in red here this is layer four and then of course we had the electrophysiology so we could look at the gamma power gamma relative power the beta relative power and the point at which they cross over actually we determined was pretty consistent in layer four and and of course then we also mapped on the other features of these things to the layers and so here's here's what we found is that the crossover the point at which the gamma profile in red and the beta profile and blue the point at which they cross over at which their powers are equal that tended to land in layer four the peak gamma power tended to land in layer two or three and the peak beta power tended to land in layers five or six and so we mapped these distances between these different electrophysiological signatures or or distinct profiles to the different layers and we did this for several different probes and for several different animals until we were able to get a very nice distribution here which confirmed that indeed the gamma beta crossover is consistent with being at laminar position zero we called the middle of layer four whereas this gamma is in at the boundary between

layer 2/3 and beta at the boundary between layer 5 and 6. and so here is our current uh toy model of what we think is going on so we think that um that there's different inhibitory neurons that are interacting with their excitatory self-partners to generate fast frequency gamma activity in The Superficial layers and in turn that there is enhance that there's another type of interneuron that is generating with slower time constants this Alpha Beta rhythmic activity so in order to examine this we again turn to Anatomy we we again turn to this question well then which exact inhibitory neurons are most prominent therefore from the layers that Express these different frequencies and in order to perform this work what we did is we Quantified the laminar expression of three distinct interneuron subtypes that are known to account for over 90 percent of the inhibitory interneurons in the primate brain and those inhibitory neuron subtypes are parvalbumin positive Cal retin and Cal bendin positive and importantly previous computational modeling work had really Sigma singled out the part of albumin cell class as the critical class for generating these gamma oscillations because of their fast time constants and strong synapses onto the cell bodies and excitatory neurons and so again we performed this painstaking histological analysis where um you you perform different stains mounting cover slipping microscopy semi-automated thresholding and cell counting in order to be able to then really have a quantitative view on which layers have which cell types that are most associated with with these frequencies and so here's what this data then looks like we have the gamma and Alpha Beta electrophysiology here and then at the same time we have the neural the presence of different neuron types the with the three inhibitory cell classes here and and then we can perform an analysis to determine which of these cell classes is most strongly associated with again and here's what we found that in in both the low gamma and the high gamma across over uh 14 regions what we found quite quite significantly was that the it was really the part of albumin cell counts shown here as a function of laminar depth that we're giving us the best explanation for the laminar profile of the gamma rhythmic activity both when you take a very wide band as well as when you perform a low versus high gamma split it comes out in both ways so how can this inform computational models of oscillations well here's what we think is going on then we collaborated with several people here on this work including um Sanchez Toto at all and uh and computational modelers in Spain and what we did was We performed an analysis where we took uh different and different interneurons with different types of connectivity so these PV these parvalbumin interneurons were modeled as having strong inhibitory connections to the pyramidal cell class and strong self-inhibition which has been important for gamma oscillations previously and when you hook them up in

these particular ways sure enough um this and you measure the spectral output of these neurons these P2 neurons superficial pyramidal neurons sure enough the express gamma and when you put in different types of interneurons with slower time constants and different types of connections to these deep layer neurons sure enough they come out with more Alpha Beta oscillations so indeed um they're they're this anatomical word could produce even more in the future even more accurate models of how these interneurons work together and with pyramidal cells across different layers in order to express these oscillations so this is really a work in progress and the the one of the first steps towards this and our ultimate aim is to really pull together this Rich anatomical and neurophysiological data together with computational models to be able to approximate the laminar structure of Cortex to a better and better degree so now so so the this next part here I'll talk a little bit about a causal test now of the predictive routing model with um with propofol anesthesia so this will be the last part of the talk here um and so we were interested in anesthesia I got interested in anesthesia actually um through my work with and contact with Jake Donahue over here who was a very talented MD PhD student at MIT at the time um and uh was was being co-mentored by Emery Brown and uh Earl Miller who was also my mentor and so we were we became interested in this topic for the following reason well first of all uh as surprising as this may seem if you go into uh to a Surgical Clinic and you're administered an anesthetic such as propofol well it turns out that the anesthesiologists may not know exactly the cortical mechanisms and the systems level mechanisms for how that anesthetic is being is actually working in the brain and it's actually to this day remains uh more often than not the case that there's actually actually no intra-operative monitoring of brain State it's it's far and few between the anesthesiologists that are that are actually doing that and so um Emery Brown who's an anesthesiologist himself is really pushing for more and more uh adoption of tool of simple tools like EEG and and spectral estimation methods by uh by the clinic and so our work was clinically motivated but it was also motivated by the ability to test this predictive routing model when um when activity especially in prefrontal cortex areas is relatively silenced and so in these experiments we administered propofol to again these Reese's macaques and we recorded brain activity there's these local field potentials and spiking activity in the awake state as well as in the unconscious State and what we first noticed is that in the unconscious State there were these slow rhythmic bouts of activity get about one Hertz one cycle per second or slightly little little faster a little slower than that those are the slow frequencies and these slow frequencies organize these massive bouts of spiking which could sometimes be very synchronous across cortex um but

that there were long periods of Silence in between these ballots of spiking and the unconscious state and so propofol anesthesia we found is accompanied by again these strong increases in slow frequency activity which create alternating on and off periods for spiking there's reduced spiking activity throughout the network but the strongest reductions are in prefrontal cortex and sensory areas are relatively spared so they're still quite reduced in their excitability in the unconscious state but prefrontal cortex is more inhibited there is also we found reductions in Cross areas synchrony and very pronounced reductions in the alpha beta band and so as a result we thought we could actually use this model this propofol anesthetic model in order to test certain aspects of predictive routing because what what's going on here we have lower order cortex that's hypothesized to feed forward prediction errors the higher order cortex and in and in turn this these higher order areas might feed back their predictions but as a result of propofol these higher order areas are profoundly impaired and very reduced in their spiking activity and the alpha beta rhythm is also the synchrony between areas is is highly reduced so what's going to happen in this case well if you think from the point of view of Simply everything is going down then sensory areas in a response to an unpredictable stimulus should also just go down but if you think from a predictive coding or a predictive routing point of view the function of these higher order areas is to create feedback predictions that inhibit activity so in the absence of those feedback inhibition inhibitory activities you might actually hypothesize that there would be an enhancement of the feed forward response when that feed forward response is characterized by an unpredictable stimulus or prediction error and so we saw we sought to test this out using the simple sensory prediction Paradigm where whoops we have a tone stimuli these tone stimuli could either be repeated or occasionally there could be an oddball stimulus and what we would expect is that there would be enhanced feedback predictions when stimuli were repeatable and predicted and that the occasion of an oddball would signal an enhanced gamma feed forward signal so we performed the auditory stimulation task both in the awake state as well as after there was an administration of propofol leading to loss of consciousness and here's what we found in the local state just as we found in the visual cortex for an unpredictable visual stimulus as as in the visual cortex so in the sensory cortex my PhD student Sophie Shang who performed this analysis found an enhanced gamma so right at the time of the local oddballs we have time here on the x-axis frequency on the y-axis and the spectrogram hot colors mean more activity to an unpredictable stimulus so there was an enhanced gamma response to this unpredictable stimulus decreased Alpha Beta to the unpredictable stimulus or said another way when the stimulus was predicted

there was more Alpha Beta And what here's what happened in the anesthetized state when we performed this exact same contrast we lost the predictable Alpha Beta signal and instead everything was enhanced so there was a decrease in frequency towards the low gamma range so ever so the the cortex just essentially went into a constant ringing in the gamma frequency range which uh to our surprise was much stronger in power compared to the gamma that we saw being enhanced during um during an unpredictable stimulus in the awake state and the way we interpret this is that in the absence of this uh inhibitory predictive Alpha Beta signal this gamma signal becomes uncontrolled and enhances itself in in the local anesthetized state here in area TPT is a sensory area you should have mentioned that so now to conclude we propose predictive routing so predictive routing is a neurophysiologically inspired implementation of predictive coding which stresses the importance of these Alpha Beta signals for implementing predictions and inhibiting sensory areas and areas of the brain that are experiencing a predictable stimulus in the absence of that you get an enhanced forward gamma output and this uh this distinction between gamma and beta across layers we found was ubiquitous across cortex really suggesting that something like this may in fact be a truly economical circuit Motif that's repeated across the brain um and and these Alpha Beta rhythmic uh predictive preparation or inhibition we found we observed was lost in the unconscious state with unconsciousness being mediated by propofol so by the way it's one of the most often used anesthetics and that as a result of this loss of predictive Alpha Beta there was an enhanced gamma modulation to these unpredictable inputs in other words the sensory cortex was in a disinhibited state and so with that I'd like to thank my many collaborators and the bosso's lab at Vanderbilt as well as collaborators on the anatomy and collaborators at MIT as well as my financial support and lastly we just started the lab about two years ago we're still in recruiting stage so you guys can check out my website my ex Twitter feed and uh and email me if interested in contributing to this line of research research and so with that I'll turn it over in case there are any questions and thank you for your attention awesome thank you Andre I'll just ask one quick question from the chat Dave Douglas asks does propofol change the activity of neuronal microtubules in any interesting way thanks Andre I'll just ask one question from the chat Dave Douglas asks does propofol change the activity of neuronal microtubes in any interesting way um I haven't looked into the microtubule kind of hypothesis that's one of the hypotheses advocated for for many years by people like Stuart hammerhoff and um there has been some recent work done by uh by George mashour I believe and so you should go check out some of his papers on on that topic but I

haven't looked into that and I am I remain somewhat skeptical to the quantile explanation uh for for Consciousness but mostly because the mechanism doesn't appear at first glance to be very biologically reasonable and uh and also that there's not a lot of there's essentially not very much data on it but that may change um in in the coming years so something to look out for great one one other question more generally how do you see experimental work in the active inference ecosystem and how do you see the synthesis of theoretical and in principle work with this kind of real wet lab and real recordings on the neurons type analysis that you're doing here yeah so it's a great it's a great question I think that what we need to be doing as a community is having more interactions between theorists and experimentalists and so we wouldn't have performed for example this experiment here if we hadn't already been inspired by uh biological and theoretical models of predictive coding and uh and of active inference as well and so that's really what we need and it is to make not only hypotheses that are turned into great theoretical and computational infrastructures and models but we also need to be thinking about if we care about the biology and we care about trying to figure out the brain and we should also be trying to map those things onto neurobiology and psychology right and so that that involves them both considering this neurobio the neurobiological elements that may Implement and mediate active inference in the brain as well as the psychological and cognitive constructs that we think are going to be particularly sensitive to capture certain elements of the of the theory and so I remain very um very welcoming to to that type of of suggestion and uh and in fact there's going to be a postdoc joining the joining the lab soon Eli sinesh who has been working at in very you know very deeply in the active inference community and so we'll have an opportunity there to both work on Theory guided data analysis but also perform more theoretically guided experimental designs that can go further into to into the neurobiological implementation and so this this was here what I've presented um really simply the first and small somewhat step towards um linking the neurobiological implementation level with the algorithmic and computational level of predictive coding and active inference and so the more interactions that we can have between the communities the better that that will that will be I think what would be um it would be somewhat of a shame is if the active inference Community went way too far into just pure Theory and that and and delved into increasingly increasing areas that were divorced from from the biology so that's of course one of the great um critiques right of the of the deep learning and more machine learning purely machine learning driven approaches uh is that they have been um you know they took a little bit of biological inspiration but then um you know

started using algorithms like back prop that were very biologically unrealistic um and so of course active inference is a response to that in certain ways but in order to continue being that I think that there needs to be indeed a tight dialogue between um experimentalists and theorists and so if you have any particular ideas I'd be I'd be happy to hear them please reach out to me um by email or um or through through X which you can find here on the slide that's that's really cool that's awesome you're gonna be working with Eli and one final question this is from Alexandra is it possible to detect the contribution of glia to the neural firing rate of the field potentials that is a question that I have not considered yet so we normally do not consider glia to be um uh active in the network but of course um that that is likely to be an oversimplification we know that they can have interactions with neurons we know that they can also have different levels of excitability um and of course that's how they can regulate neurons metabolically but as far as the contributions of glia directly to the local field potential I am not aware of how to uh how to determine whether or not they make a significant contribution um the I actually haven't seen any any work on that but thank you for the question it's uh it's it's always good to read reconsider our core assumptions right about the the types of the cell types that are most important to to capture so it's certainly something to think about thank you thank you all right thank you Andre super cool work and good luck with everything until next time bye -bye all right next up is going to be Keith Duggar who has submitted a pre-recorded video the pre-recorded video is called active inference and the actor model so I will bring that up right now foreign Keith's pre-recorded video on active active entrance and the actor model hello hello I'm Dr Keith thutter platform CTO at x-ray an extended reality AI company and co-host of machine learning Street Talk podcast now as we all work to build an ecosystem based on active inference software will obviously play a foundational role to make the most of the active inference we'll need to use software engineering paradigms that align with the principles of active inference and I think there is one tailor-made for our needs it's called the actor model active inference in the actor model are two deeply connected understandings of the world they provide foundational Frameworks for dealing with the Dynamics of complex systems with a focus on autonomous agents that interact in an Ecology of nested systems I'd like to for some of their key connections including the role of Agents concurrency autonomy uncertainty and behavior adaptation we will see that active inference in the active model are both Paradigm shifts away from a deterministic centralized step-by-step thinking to a decentralized network concurrent perspective of both computation and cognition just a little bit of history about the actor model back in 1973 Carl Hewitt Peter bishop and

Richard Steiner were all working at the Massachusetts Institute of Technology AI lab to fundamentalize a concept of concurrent computation that included both structure as well as adaptable algorithm execution conventional Methods at the time lacked robustness and secure mathematical Foundation there are collaborative effort ultimately led to the creation of the at the time it was viewed as revolutionary due to its characteristics of both increased error tolerance and distributed computation abilities throughout the 1980s and the 1990s the actor model became the basis for numerous research projects as well as practical projects gaining popularity for its flexibility and intuitive approach to concurrent computation it was primarily used in artificial intelligence and multi-agent systems sound familiar new after-based languages like actor salsan and erlang contributed to the refinement of the model shaping it into a more robust and flexible approach to concurrent Computing and it remains alive and well in computer science today more recently the atro model is getting renewed interests mainly due to the growing need for distributed systems cloud computing and Edge Computing fueling the internet of things and web 3.0 these computer tasks are ideally suited to take advantage of the aqua model's architecture which is designed exactly for modeling concurrent handling of those large volumes of data on the one hand and fine-grained disparate autonomous systems on the other hand this application of the actor model has had profound effects on major companies that have utilize its principles to handle big data problems such as Twitter Facebook LinkedIn so what does this have to do with active influence I'm guessing you've already heard some of the parallels in the intro well let's start by looking at a few of core principles of the actor model and how they relate to the principles of active inputs let's start with the concept of isolation isolation means that an after in the actual model does not share its state with any other actor it can only be affected by receiving a message and it can only effect change in the state of other actors by sending a finite number of messages in response from a software engineering point of view this isolation principle limits potential side effects of an operation to a single actor thus improving the system's overall predictability the liability and most importantly if embraced fully can actually simplify the design looking at the diagram we've seen ecosystem of actors sending messages to a particular actor which in turn sends back messages to other actors where is active inference well let's recast receiving and sending messages to a perception action cycle and denote external internal sensory and active States and we now clearly have the necessary Foundation of activation active inference a markup blanket the actors of the actor model map directly to the Agents of active in addition the finiteness the fact that an actor can only send a finite number of messages in response is

also an important share of property because active influence models reality it necessarily respects the resource constraints of real systems and this is nicely bathed into the foundation of the OCTA model let's look at another core principle asynchronous message passing communication between actors is asynchronous this means an actor doesn't wait for a response after sending a message it continues working it continues living as it were this is critical as it decouples the actors leading to A system that can continue functioning living and making progress even when parts of the system are slow or even temporarily unavailable Professor friston has said that the free energy principle is the ultimate existential question if things exist what must they do well the actor model claims they must not wait on others of course an actor can choose to wait on others but it must not be forced to do so in the model it must be free to choose this leads us to another critical principle that both models share autonomy the free energy principle is a model of physical reality and our reality is after all concurrent all throughout Infinite Space Systems are evolving simultaneously according to their local Dynamics and therefore this is reflected at the heart of the free energy principle of course a model of computation need not constrain itself to physics but he would at all were seeking to develop a model that did model the reality of distributed concurrent systems and luckily for us the actor model Embraces both concurrency a scene from The principle of isolation and after autonomy making it compatible with active next we come to nesting the actor model allows for an actor to not only receive and send a finite number of messages to perceive and act it also allows as an action the creation of a finite number of new actors these actors can either be nested within the parent say the parts of an animal cell or it can be released into the environment as independent actors from then on this principled model fits nicely with the beautiful concepts of multi-scale nesting and active inference this allows for actors to contain ecosystems of actors both all the way down and all the way up lastly I want to cover two more active model design principles Behavior change and persistence actors have the ability to change their behavior in response to a message this adaptability allows for the construction of complex stateful entities that can evolve over time and indeed it allows for an entire ecosystems to evolve new emergent behaviors when used for software engineering decides a powerful tool for managing complex Dynamic systems active inference of course Embraces this to the extreme very essence of thingness is the ongoing attempt to predict and add to an environment and thereby continue to exist to maintain one's mark on blanket in a broiling sea of activity along with this also comes the concept of persistence persistence allows actors to save their state and to restore or modify it later a feature that embodies principle memory

memory is a prerequisite for Learning and adaptation an agent's ability to predict depends on its ability to remember past experiences and thus minimize the surprise associated with unexpectedness the vital role of memory is also emphasized when we assume that agents have and don't deprioritize either from experience or inheritance contributing to their world model this world model both drives their current behavior and it is continuously updated based on new experiences contributing to their ongoing adaptation and existence okay great you say there are clear and deep connections between the actor model and active inference but how does this help us in the active inference community well firstly in my opinion it is a software engineering Paradigm we should embrace and if we do there are of course after model libraries and Frameworks that we can use such as akka Orleans thespian tactics Proto actor and many more which we can immediately use when building active inference software modules and applications there are also libraries languages and you know language features that align very well with the actor model principles such as zero and Q Tokyo and rust erlang async await and C sharp Etc but more important than the tools available to us today is the software design mindset that will guide our creation of the active inference software of tomorrow the actor model provides a paradigm of software design and Engineering that is the most perfect match that we have for active inference this is evident not only from the alignment of the core principles we previously discussed but also from the insights that active inference in the actor model bring to each other for example consider what is now called Hewitt's law informally stated as everything is everywhere this law signifies the idea that in a truly asynchronous distributed system it can take an arbitrary amount of time for a message to go from one place to another and any actor must be prepared for that event there are simply no such thing as instantaneous in such a system and no component can make an assumption about the timing of another component's actions in fact one must act as if a message may never arrive this has important implications it's just that it is impossible to accurately and consistently determine the state of the entire system at any given time because the information just may not have even propagated across the system and also attempts to implement Global synchronization will inevitably introduce bottlenecks and reduce efficiency Hughes law emphasizes the need for systems to be designed in a way that they can effectively handle these unavoidable delays and uncertainties highlighting the importance of robust non-blocking communication mechanisms and local decision-making abilities in short a Hector model systems are by Nature non-deterministic does this sound familiar what other Paradigm highlights operation under uncertainty in the autonomy to continue despite the environment

Act of inference in the free energy principle active instruments reflects the reality of an unpredictable World in which our software systems operate different outcomes may result from the same initial conditions due to the occurrence of events in a random unpredictable order this is the concept of surprise that we all know well where an agent updates its beliefs about when the sensory input it receives does not match its predictions both the actor model and active inference acknowledge that the world is unpredictable even more than acknowledge it the models accept this uncertainty as a given and not something to be managed away indeed as We Know in the free energy principle the uncertainty that we maintain in our models is what gives us the flexibility to adapt perhaps this is just my personal line of fancy what I imagine a future where software modules Guided by active do away with hard-coded error handling and swap in probabilistic learning algorithms that optimize themselves as the Aero landscape evolves modules that are robust and self-healing distributed systems with no single points of failure that focus on predicted disaster avoidance rather than reactive disaster recovery looking towards the future we as a community have the potential to push the boundary of both active inference Theory and practical implementations of the active by leveraging the strengths of these two paradigms together we can create software systems that are robust adaptive and more aligned to the physical world in which they actually operate imagine the future where software components using active inference in the actor model can anticipate potential issues learn from past mistakes and adapt in real time to environmental changes with this approach we can build systems that are fundamentally more resilient and more efficient in my opinion this can bring a step change in software reliability and performance and scalability and Heralds a new era of computing weaving principles of biology and cognition into the fabric of our software systems bringing them closer to life in the process in conclusion the coupling of active inference to the actor model provides a powerful new lens through which we can look at software design and engineering whether we leverage existing languages and libraries aligned with active or invent new ones we are standing on the break of of an exciting Frontier so let's seize the day have a look at the actor model and its relationship to active inference and let's shape the future of intelligent distributed computing thank you for listening awesome great talk by Keith thank you Keith for sending that in there were some comments in the chat so Keith potentially if you want to join for a q a some future time but really cool presentation all right the next presentation is by Sanjeev namjoshi and this presentation is going to be called developing next-gen active inference tools broadening accessibility educational resources and the software ecosystem I'm

going to start this talk right now hello everyone and thank you for being here my name is Sanjeev namjoshi I'm just going to restart it machine learning engineer working at the AI Services firm just because it's a little quiet all right restarting the talk here my name is Sanjeev namjoshi and I'm a machine learning engineer working at the AI Services firm Kung Fu AI where I primarily focus on computer vision projects today I'm going to be talking to you about my progress toward providing greater accessibility visibility and knowledge of active inference and the I'm excited to be presenting this material at the Active inference Symposium this year because the idea of an enacted ecosystem of shared intelligence perfectly captures the philosophy that underlies the projects I'm currently engaged with I've spent the last seven months on sabbatical for my job to work exclusively on an active inference textbook and related tools presenting chapter presentations code reviews and receiving feedback weekly at the Active inference Institute The Institute has provided a space where interdisciplinary research can flourish as the connections and influences of active inference spread to other fields it is consistently fostered the spirit of collaboration and shared intelligence that I wish to embody in my own work as part of this ecosystem I intend to continue closely working with the Institute to provide materials that will bring active inference to a much wider readership I originally chose this project when I saw the great potential in the active inference field and couldn't help but make a comparison to the state of deep learning in 2006 when neural networks or one of just many possible models rather than the dominant choice in Academia and an industry this of course all changed in 2006 when Hinton and his colleagues released the Deep belief Network paper which is generally understood as a start of deep learning as we understand it today after some Hardware Innovations and the release of the well-known imagenet Library we started to see coverage and successive AI in the news as well as in academic research but in 2012 deep learning truly provided its value with alexnet and for the first time deep learning achieved better than human performance on image detection tasks start What followed was a proliferation of deep learning all across industry and research I've added in some well-known Milestones just to highlight the explosion of progress in deep learning in the last decade though there is so much more here that we could discuss so what about the state of active inference as a field from my perspective active inference lies in the same position as deep learning in 2006 influential and on the brink of exploding in popularity this paper which is from contributors at the Active inference Institute shows the current growth of Publications in The Institute and its community in active in the last three years the active inference field has seen a number of important milestones here I show just a few that broaden

the scope and attention to the field we have the first International workshop on active inference in 2020 we had the first active inference Symposium and the founding of the active inference Institute then called the active inference lab we had the release of the par at all 2022 textbook and the pi mdp python package and I see myself now as perfectly poised to bring active inference to Greater visibility and attention this is in part because of the current academic interest and deep reinforcement learning and generative modeling working alongside the institution and other organizations my aim here is to provide some of the fundamental materials to capture the attention and notice of machine learning researchers and students to bridge this Gap to bring active inference into its Renaissance to this end I've been working for the past seven months on sabbatical to finish work on a comprehensive textbook the aim of the textbook is to provide the tools to bring active inference to a wider audience primarily those in machine learning research and applied Fields such as Robotics and to decrease the challenge in learning the material largely by separating it from much of the Neuroscience background that is usually a prerequisite this decrease in prerequisites means Labs will have to spend less time helping students becoming acquainted with the field and researchers outside of Neuroscience will find this book an accessible entry point that uses terminology familiar to machine learning rather than Neuroscience fmri image analysis all derivations are in one place currently in the field many derivations are spread across different papers even behavioral papers instead of just technical ones and it's hard to know where to look if you want to understand a particular equation or concept part of the success of deep learning in the last decade has come directly from focusing on narrow improvements to specific aspects of the modeling there are many open questions and areas of research such as how to prune policy trees exploring second order optimization rules for state and parameter updates and scaling active The increased accessibility for researchers would also lead to many new industry applications such as autonomous vehicles robotics video game design and AI the textbook would also put a spotlight on Bayesian mechanics and invite contributors and contributions from researchers as an exciting nascent field grows and develops part 4 is largely a literature review and can be very helpful to those writing about active inference from Fields such as philosophy psychology sociology and many others and the historical context sections that are part of this book provide a lot of that context as active inference is built upon Decades of research in Neuroscience psychology and many other fields and also draws upon current work in many fields that have emerged in the last 25 to 30 years finally latex reproducibility may offer interesting ways to rearrange the book and integrate it with the code for an online only

experience now I'd like to share with you some of the progress of my textbook in the general structure the textbook is divided into four parts the first part introduces fundamental concepts to set the stage in particular I have focused here on presenting well-known statistical ideas from the perspective of an agent modeling its environment who States it must infer from an observed noisy signal the second part focuses on continuous and discrete State space formulations of active inference where the algorithm of focus for the continuous State space formulation is active generalized filtering part 3 which I'll begin writing in a couple of months focuses on a sketch of Bayesian mechanics and the required background designed with the knowledge this field is still dynamically changing and evolving here I will focus on some of the fundamental concepts and ideas as well as code simulations to allow readers to get a deeper and more intuitive understanding of some of these challenging ideas finally part 4 is a systematic literature review that covers all the various applications and extensions to active inference that have been innovated in the last six to eight years these applications include things like robotics all the behavioral modeling and neuropsychiatry and human and animal behavior theory of mind and so on the extensions talk about how applications of active inference can be used to talk about Dynamic systems more generally and apply to things like ecosystems and to economies and other types of things like governance and so on as of this week the rough drafts of part one and two are complete and 10 chapters have been presented to the active in support of this textbook are four separate tools that I will discuss over the next few slides but before I do that I would like to First highlight some special aspects and features of my approach to this textbook the major focus is on writing this book for a machine learning audience or students learning in this and adjacent areas Neuroscience is out of scope for this book many of the recommended and even optional prerequisites that are shown here are typically known by undergraduate students in science and engineering and certainly by graduate students in these fields nonetheless the book is written to be readable by those that desire to focus on everything that is the math the code the concepts those that just want to focus on the math but are not interested in implementation and even those that may skim over the math and just try to understand the ideas intuitively one thing that's very important to me in trying to express these ideas clearly is by spending a lot of time working on typesetting and style which is very important to successful learning so I've spent a lot of time attempting to make my work clear and readable to this end I have margins which collect specific key terms for references later which you can see in some of these figures that are shown here and these terms will eventually correspond to the ontology project ongoing at the Active inference

Institute margins also provide further explanation to accompany the text and this will be useful to readers who want more detail and explanation there's a large focus on building an intuitive understanding of the concepts for example importantly all algorithms are explained from scratch and this book we typically start with a description of the agent environment modeling problem we then start the book with a univariate case extend the multivariate case then we introduce variational inference we add Dynamics generalize coordinates where applicable hierarchical models action and also learning and other modifications we might add to our models we have a very big focus on figures clearly Walking The Reader through the text and giving detailed visualizations of important Concepts and in terms of how the textbook is set up the early part of the book focuses on basic concepts such as hidden State estimation that is estimating the conditional distribution of a latent variable given some observed data the aim here is to explain the modeling Paradigm in the context of an agent attempting to infer the states of the environment that is the interaction between a generative model and a generative process a perspective that differs from the Bayesian inference style that is normally taught um in universities and and many introductory textbooks typically introductory textbooks on Bayesian inference focus on parameter estimation or learning and part one introduces the expectation maximization algorithm as a way to explain the connection and separation between hidden State inference on the one hand and parameter learning on the other additionally a large Focus has been placed on variational inference which is explained in detail and the book features a catalog of all the different forms of variational free energy and expected free energy in the literature and how they can all be derived from one another the book also covers raw and Bollard style predictive coding terms and ideas such as key ideas such as prediction error minimization as well as clear and intuitive explanations of fundamental concepts such as surprisal the textbook focuses heavily on building intuition through derivation and the general flow of most of the chapters is to set up the problem that needs to be solved which is defining an interaction between the agent and the environment showing the elements needed to solve it which is usually random variables and parameters that form a joint distribution or generative model replacing probability distributions with their algebraic formulations and then moving through the algebra to a final analytic or gradient based equation the reader should be able to recognize most equations in the literature upon reading this book I'm also making extensive usage of Bayesian networks and other types and styles of graphical models such as Factor graphs will appear in part four there are hundreds of custom figures that have been created so far and the figures are detailed to give the readers a

deep understanding of the different types of content that is covered throughout the books and also summarizes much of the information and equations that are pervasive throughout the active inference literature another important focus of the textbook is that many of the models that are presented are also shown in pseudocode which should Aid the reader in implementation and finally each chapter is filled with numerous what I call experiments and these experiments correspond to the Jupiter notebook and try to show the application of a concept in a simulated so a lot of these experiments start out by generating data so we have some kind of generative process and then we have that data passed to a generative model or the agent which then attempts to either perceive and learn from that data and even act on it and the example that's shown on the right margin here this is just a perception problem on a continuous grid um which has been divided into pieces for the purpose of the simulation but it represents a continuous State space and the agent in the bottom left corner shown as a mouse as a prior belief about where some reward food is and its environment but it then needs to use perceived from sensory data that it observes the true location of that reward or food which is obscured or occluded in some way by the Mist that's shown in that figure so these types of experiments give the reader a better sense of how to apply these statistical ideas to a real world situation so we can understand how it might apply to some kind of autonomous agent next I'd like to cover and shift my attention toward Jupiter notebooks and videos and like to note that upon publication of the textbook these Jupiter notebooks will be released on GitHub and should be fully reproducible using Docker and other version handling tools one of the big emphasis is on the Jupiter notebooks is it has to be direct correspondence between the equations and explanations in the code and in the text this will build a direct understanding and show applications of the concepts explained notebooks are filled with simulations and visualizations many that appear in the main text and in addition to that I have also over the last seven months been presenting chapter presentations to the active inference Institute so far a draft version of 10 of the 10 chapter first 10 chapters of the book have been recorded at the Active inference Institute and these are just some sample slides that I prepared that try to explain these Concepts in great detail in the final stages of writing this textbook video lectures will be re-recorded and released alongside the book I also plan to create detailed code walkthroughs videos that walk through the different examples in the Jupiter notebooks now I'd like to talk about a few planned future resources I'd like to work on after the book is complete or toward the final stages of the book to have further support and Educational Tools some of these planned future resources include a software suite in Python to enable an alternative

learning approach for those who do not wish to learn about the algorithms from scratch and this will expand the possible landscape of Engagement as Pi mdp already exists I will not be working on a discrete State space python package but I'd like to fill the space for things like active generalized filtering and also just an availability of different types of simulations of Bayesian mechanics as it's currently defined today or at least the different versions and varieties of some of those key Concepts I'm also very much interested in Interactive Learning and my aim would be to have these preset simulations Concepts that are explained in text with various simulations interspersed in the form of plots and demos and other visualizations and the idea would be that the user could manipulate sliders and knobs to tweak various parameters that would help aid in learning as they get a feel for how these systems behave especially ones most of these systems we talk about are dynamics so seeing how they change over foreign the active inference Institute has made tremendous progress in the past couple of years to provide a collaborative environment of res for researchers and for students of active I hope to be part of this ecosystem as I continue to support the spirit of accessibility and collaboration and I'm excited to continue to contribute to this ecosystem of shared intelligence and look forward to what we can build together I would like to thank the active inference Institute for hosting my presentations code reviews and feedback sessions and inviting me to present the Symposium and also thank my employer Kung Fu AI for letting me take time off to write for the past seven months please feel free to contact me at any time email is the easiest way but I'm also available on the active inference Institute Discord if you'd like access to the textbook and related materials please send an email requesting access and I can get you set up and that's all I have for today thank you very much awesome all right thank you Sanjeev greetings welcome hi Daniel thank you yes well we're really looking forward to your presentation you can share your slides and go for it great can you see it uh Yep looks great thank you all right good so thank you so much for for having me I'm really happy to present this paper today and particularly because it's the very first time that I present this work so I'm really looking forward to um any um feedback um and any reactions um to these work okay so um the aim of this paper was to apply the free energy principle to analyze the interdependent Dynamics between living systems on the one hand and non-living planetary processes and this hopefully would we hope allow us to gain some insights into the future states of Gaia and by Gaia we're going to mean the Earth and formulate some predictions and devise potential interventions to alleviate future dysfunction on the planet okay so then um line up for the presentation is I'm first gonna go over in brief um the theory uh the concept of uh Gaia so

known as the guy hypothesis then I'm gonna go into just giving some Essentials on the financial principle which um I'm assuming that um it's not really necessary but uh just for the sake of the flow of what I'm my argument then I'm just gonna present some things on the free energy principle and specifically on self-organizing interacting systems and then I'm going to apply the free energy principle to the guy hypothesis in a sort of like comparison analysis is in like how do they if at all complement each other for epistemic gain and then um I'm going to after uh having done that this should be um should allow us to inform or to gain some insights as per the human behavior in relation to Gaia or the planet Earth okay so let's see how it goes so first I want to introduce the notion of biophilia which plays a fundamental role in this work so biophilia came first um or came more prominently in this particular book here and it's quite interesting um the idea is that the relation of humanity to the rest of the natural world is essentially one of dependence and this is very much already in the literature free energy principle um but more than that there is this uh view of the necessity of a continued survival and flourishing that is utterly contingent upon the state and functioning of the environment so we're talking about things like very utilitarian things like survival flourishing mental health right so individuals interact with the environment so to achieve that but then on top of that we wanted to add another conceptual layer which is or concerns how humans possess an innate cognitive disposition towards nature which is known and developed in in this particular other field of work as biophilia and this is the idea of a natural psychological disposition of humans to seek out and connect with other living organisms and the natural environments so this kind of like the idea that is encapsulated in the whole framework and Theory under biophilia so then um this seems to if that's the case of as things are described by biophilia then that seems to um to bring us to a dilemma on the one hand we have these natural tendency so to speak of humans to engage with the natural world in a very uh positive and constructive way and on the other hand we see the lack of action to protect the Earth and here we have in mind particularly the case of the climate crisis so in the presentation I'm gonna make my way slowly towards getting to that and um to better understand the Dilemma okay so let's go uh in brief over the Gaia hypothesis so this is a a concept that comes more prominently in James Lovelock's book Gaia a new look at life on earth and the idea there is that 3.8 billion years that life has persisted on Earth and conditions have remained remarkably constant and unfavorable across a huge range of parameters the guy hypothesis is that this consistency or constancy to be is a natural product of evolution in a modification however of the canonical Darwinism uh the guy hypothesis somewhat modifies canonical Darwinism how well by thinking that rather than

evolution of the individual organisms on an inert environment what divorce is the whole earth system with its living and non-living parts existing as a tight-coupled entity so this is the kind of like so to speak refreshing or new um way of thinking that somewhat reformulated a little bit of Darwinism right it's not the emphasis is not how one's individual organism evolves or species evolves but the whole earth as a system so that's quite interesting because for example um the the view that the combined genetic wealth of information of the whole system becomes much more prominent here than the darwinistic kind of view and this acquires characteristics there are conducive to its continued existence and in turn affect changes on the environment so then seen as a whole organism or as a whole system one can look at Gaia or the Earth as a complex system which appears to have the goal of regulating the climate and the chemistry at a comfortable state of life and these views and these ideas have come to gain a more visual expression in 2001 with the Amsterdam declaration on Earth system science precisely endorsing this view on the planet Earth as a whole system so then the guy hypothesis is a complex system in in in various ways because it changes adaptive in an Adaptive and non-nonlinear way it also displays dynamical image emergent and sudden tipping behavior and when bifurcation points are reached the new attractive states form new attractive states are formed and this means that it also self-organizes and there is interdependence between the processes of the system and finally the guy hypothesis seems to be quite interesting and extraordinary in the sense that it brings together so many different fields such as ecology climate science cybernetics complex systems theory chaos mathematics and philosophy of mine and that's just a few all right so then um the idea is of the guy hypothesis is that we could look at the planet from orbit or as it is known the all review effect and the guy hypothesis offers this glimpse of fundamentally different way of thinking of the planet of thinking that in fact dominated most mostly by human history so it's a little bit of like um that um very inspiring clip by Carl Sagan the pale blue dot whereby the world is considered so throughout the human history there's typically a tendency for the world or the planet to be considered from a single living organism though rather than an it so that's that's the idea of the overview if I can take taking this overview effect look at the planet as this whole living system rather than it okay so then we can think about the guy I apologies as like Newtonian physics or even as the free energy principle the guy hypothesis can be seen much more as a framework or a method of Investigation rather than a falsifiable empirical claim so in this sense this particular Frameworks they are and should be judged by their usefulness and not by their purported correctness where usefulness importantly usefulness is to be determined by factors such as

parsimony coherence and consistency plus its empirical and epistemic power to explain phenomena produce new facts and so on in a given context and at a given time frame so that's the stance on that we think that should be taken on the Gaia hypothesis all right so now having said a few hours on the guy apologies now um I'm gonna just prepare the stage by saying a few words on the financial principle which are given the topic of uh this conference needs an introduction but for these purposes I must mention just a few things so the free energy principle we take it to be the case as well that it is a methodology to understand um over-served patterns of behavior uh sorry observed patterns of behavior in the natural world so then we apply the free energy Principle as the tool in methodology to analyze these interdependent Dynamics between living systems and planetary processes right kind of like these goggles this methodological technical scientific goggles that you put on to gain insights into the future states of Gaia or the Earth such as to make future predictions and wise potential interventions in this particular case with the care of alleviating future potential or present actually dysfunction okay so the free energy principle um very much align aligns with um with this uh Thinking by Ewing uh schwelinger and the view that the second law of Thermodynamics or under the second law of Thermodynamics starts open systems like us should tend to dissipation however interestingly there are displays of this super cool thing the negentropy which are pockets of the universe where order and bodily Integrity are maintained in the face of the search towards chaos or towards entropy or towards this patient and this view of the pockets of the universe is incredibly interesting because this is where we find the self-organizing systems like us and what is so peculiar that distinguishes these self-organizing living systems from non-living systems is that they have the capacity of acting upon and exchanging matter energy or information with the environment in order to seemingly idify the second law and therefore or thereby avoid participation and chaos and entropy so that's extraordinary so then active inference um becomes this very useful tool to tell us or to explain elucidate how um these systems seemingly defy the second law how this they resist entropy how the self-organizing systems combat entropy through the interaction maintaining their integrity via homeostasis so then they maintain this energetic Integrity they utilize energy to stay within a specific range of states that is vital for their survival and preserving their existence and they also engage with these feedback Cycles which are the feedback actions uh like drinking when thirsty or moving away from heat um signify working accomplished by bound energy contrasting with the free energy the thermodynamics as I just mentioned by assumingly challenging the law they minimize entropy and Safeguard their free energy and

finally they this behavior that we observe in this self-organizing living um in striving to minimize free energy um is or agrees with a certain principle that we know which is the free energy okay so then um as I said um earlier and as we know but just just uh to keep in mind then active in France becomes a very useful tool to understand and make predictions about these coupling Dynamics taking place between the living systems internal States in both is both operation and fully closed and thermodynamically open to the external States creating this distinct boundary which we usually mathematically capture with Marco blankets and then this interaction unfolds through the military or active and sensory States then the key Point here to keep in mind is that what we want to have for adaptation and well-being is that there is a balanced interaction between sensory and active States and this is because it is in within the interaction between sensory and active states that so to speak all of the important things happen in order for the system to be well adjusted attuned adapted to the so living systems avoid this dissipation they want to remain alive they do everything in their power they navigate their environments in order to avoid this dissipation which is also or can be also said to minimize free energy and they avoid their dissipation by doing either one of the two things either by changing the model or changing the World by acting upon the world and this is extremely crucial for the point that I want to make later on so now um I want to apply the financial principle to the guy hypothesis and see how that comes together in identifying and developing and why do I want to do that I want to do that because I've also already mentioned it but I'm just gonna refresh it because I want to identify and develop interventions to rectify any potential maladaptive responses namely actions which are anti-tential to maintaining the Integrity of the system what we want is to have healthy well-adjusted adaptive systems and human behavior particularly that leads to the imbalance within the mark of blanket States is or could be considered sometimes maladaptive or pathological and this will be the last point that I'm going to make in the presentation but for now um I want to focus on how does the French principle deal with the guy hypothesis as a self-organizing system like with or just saying the guy omits the requirements to be formalized as part of an active inference system it exists in a non-equilibrium equilibrium steady state which means that its processes and states must persist over time within a range of states that are far from equilibrium that means resisting entropy and the Gaia is also distinguishable from its larger scale Cosmic environment such as other planets and the rest of the universe it is also defined by the conditional Independence that we were just talking about between the internal and external stage which influence each other only vicariously so then we can taking that into consideration we can even see that some

of the people have already done this kind of like line of reasoning they've already run it through and they did that by employing Marco blankets to analyze the earth's climate System Dynamics which is quite interesting what they did was they looked at metabolic rates of the biosphere and they represented them as internal States and space weather as external States and then they looked at active States as mirrors greenhouse effect shifts and sensory States as echoed ocean driven temperature changes and then internal and external states are the idea here is that they indirectly influenced each other through oceans delayed response then earth's climate is interpreted as anticipatory minimizing variational free energy so we take these work and we want to take it a step further um agreeing with this work and building upon this work or which shows the climate change uh we wanted to look at um the more broad understanding of Gaia through the lenses of the free energy principle so instead of focusing on climate change which is very brilliantly done by these researchers we wanted to then scale it up and apply the free energy principle to understand um the Gaia so then one can think of the internal States as the biosphere geographical processes and the external States as this everything that that is beyond the system such as the outer atmosphere celestial bodies and molten core then we can think of sensory States as the external environment in part such as the earth rotation orbit Tides affected by the Sun Moon planets plate tectonics climate weather influenced by solar radiation space weather Etc and then we can think of the active States as Earth's adaptability and adjusting properties from external influence such as gravitational force authors orbits magnetic field interacts with solar winds or emission of detectable radiation influences distant objects that's a few examples so then I want to focus on the active States active States then through this lens active States then embody its primary mechanism for self-regulation representing its efforts to remain within the bounds required for ongoing existence based on predictions about the present and future states of the system for for instance the current levels of greenhouse and orbital effects soil PH and the number of living organisms reflect the predictive attempts of the Gaia system to adapt to anticipate anticipated effects of solar events volcanic eruptions ocean salinity changes and other environmental factors in the future so it is in this context that the role of life itself assumes a Paramount importance in the self-regulation of the biosphere because the role of life itself and particularly as we are going to focus on um next the human life has a particular influence on all of these sort of already natural Dynamics going on uh in the guide um this allows us before we move into the human effects this allows us to make one final point before we move there which is to address um a criticism that has been made in this paper by vicenter haha on the mark of blanket trick and

um what they um have identified is that a purported weakness of the financial um what we say is that a weakness of the financial principle that is identified in this paper actually speaks in favor of its application to the guy hypothesis and let me just try to explain that and colleagues they rightly point out that markup linkage do not automatically capture every relevant property of biological systems relational properties such as affordances and by affordance system in what the environment offers the animal what it provides or furnishes either good or evil fall into this practice so what they're saying is that Marco blankets do not automatically capture every relevant aspect of biological system such as affordances right and we want to point out that um however affordances correspond to Gaia as one wrong in the ladder of the nested Marco blankets within the system in question at every level ranging up from say cells organs animals ecosystems Soul systems the whole Cosmos and how is it that we think that this particular framework that we bring here does have or solve or address the notion of affordances our solar system is one that affords a living planet so the affordance available is an environment in an environment are not determined solely by the physical properties of the environment and now we think that this is the the point that we would like to address in haha's paper is that the affordances are not properties of the environment but also by the organism's goals abilities and previous experiences and in this case of the Sun so what is meant here is that the sun is what has a furnaces for or what the force anything else to happen on the planet at whatever each scale right the physical properties of the Sun as well as the goals and abilities of the organisms on Earth shape the affordances that the sun provides so it's a quite interesting way of thinking that any importance is whatsoever that we might think of that we have as living systems they are afforded by the Sun so that's that's the idea okay so um now having gone through how the financial principles somewhat um speaks to um and contributes to understanding the Gaia hypotheses um then we want to focus on how human behavior has a profound effect on the Dynamics already existing in the natural world okay so according to the free energy principle living beings have a natural tendency to act upon their environment to prevent the suspicion of their integrity so that's what we do in order to remain alive there are therefore two ways for a thing like us to minimize free energy and thereby maintain our own Integrity upon evidence as we navigate the world and gather and observe evidence we have to to options one is to change the model the other one is to change the world by acting upon it like I just said before so then why is this relevant well this is relevant because there is uh the possibility that um that in ways in which living systems and in this particular case human beings interact and adapt to the world there is a possibility that we see maladaptive Behavior

and how would that be well if it is the case that there is a rejection of the information available so as to remove the imperative for action or if it is the case that possessing the information available but we still remain or refrain from taking action then there seems to be if the free energy principle is correct if the financial principle is to be correct that systems living beings tend to act upon the world in order to remain alive um and if it is the case that one rejects that kind of information and does not take action to remain alive such as the information of the climate crisis then we might be looking at maladaptive Behavior so if the free energy principle holds then the observed contradictory behavior is a maladaptive state or should be seen as a maladaptive state that arises from an imbalance within the mark of blankets and what is how do we see this imbalance within the Marco blanket we see it to put it in very very simple terms we see it as an insulation of internal states that are not being affected by the information available outside of the system so um the the influences or the interactions are occurring between internal active and sensory States and this is an imbalance that results in an insulation or an equal chamber effect in internal states that are impermeable to the information from or potentially existing in the external states that demands action so then some foot for Thought um what what can we make of this the free energy principle if we are to use uh the financial principle to model the Dynamics of the relationships between humans and the environment then we think that it is possible to see the current lack of pro-environmental behavior as a pathology of the human species that must receive treatment we label this condition as biophilia deficit deficiency syndrome biophilic deficiency syndrome and and we use we it is important for us to focus on that to learn to learn some constructed to construct some validity is useful to consider um in this case other applications of the free energy principle that may be useful as an analogous case of pathology mainly conducted in Psychiatry and neurology and in here and very very very short because we don't have time the idea is to see it as a failure to assign the right way or prediction to prediction errors so then the free energy principle um from this point of view this is a prosenius pathology because doing nothing is a base optimal response in the face of pathological attention to various sources of evidence so evidence is um ignored so concluding and and recapping according to the financial principle living systems possess a biological encoded inclination to interact with the environment in order to survive and adapt to the environment and despite the climate change being a pressing issue and the information being available the evidence being available proactive measures to reverse its effects are not giving the priority they deserve um these State of Affairs seems to show very clearly a Divergence from a natural inclination that we call biophilia or that it's existing in religious

Torres biophilia this discrepancy between the expected and The observed actions can be understood in the framework of the free energy principle or so we think we think as an illness it implies that systems including human societies may form erroneous inferences that prevent them from effectively addressing the problem at hand now thinking about future directions and this is my last slide um current ecological imbalances could be thought of as a stuck State as as that require an active disturbance to bring about some change if not a stuck state is a state that a system is at a local minimum where the system is at and won't get out of that state unless it is Disturbed right so that's that's the principle in in the complex systems theory and we think that that's how these um particular State should be looked at that's for future research and what we would be calling is we would be calling for amounts to the kind of homeostatic Awakening so the sort of like getting out of this text state would require some kind of like homeostatic Awakening A deliberately induced disruptive shift in the trajectory reflecting a prioritization of the planetary immune stasis over infinite growth or staying in the same place and yeah this is my last slide thank you so much I would like also to acknowledge my co-author Casper Montgomery um thank you so much for your attention yes thank you well a lot of ways to go I'll just ask one question just in the years that you've been in the game how do you see the octave inference ecosystem and the ways also that it's being applied to ecosystems evolving like over what time scales or in what ways does that really presents itself to you hello I think there's um there's a lot of work that needs to be done um we try to do on this particular one here to apply employ um active inference to um to the the planet or the Earth as the living system precisely to bring out or to overcome and um so so one is to to to to to to look at this problem of the climate prices as what we think it is which is more of a pathology if the Friendship principle holds then it seems that there is a maladaptive behavior of the collective of the human species um so I think that there's a lot of work that can be done and that would very much be useful in the climate crisis um in general um there is some other work that came out more recently by Maxwell and colleagues as well um on the ecosystems which is really really cool and I would also mention that one and and I really hope that we can we can pursue this this line of research and employee active inference in the French principle um to this particular to this particular case what what what really one of the things that really attracts me to the free energy principle is the capacity that it has to uh to bring to the Forefront um uh certain laws of physics but also the the living systems navigating and interacting with their environments which I find much more appealing that any computational framework per se so that's what what I I find that it's a very useful conceptual toolkit that I think that I would really

like to see more applied in the future for the climate crisis on the Earth great well thank you so much for building on that line of work and for sharing with us tonight so thank you so much for having me until next time see you thank you bye or gray talk by an S and now welcome Aspen how are you doing hi Daniel I'm good hanging out yeah yeah yeah sounds good and uh yes looking forward to your workshop on sophisticated inference in pi mdp so it can be um up to 90 minutes or it's totally okay if it's less and we can take a short break but please um take it away and just let me know however I can help thank you so much um maybe I'll share my screen and start so regarding the structure of what we are doing here um I'm assuming that people who see this later might want to try it Hands-On along with the tutorial or something so that's the structure I kept in mind and so I'll go slow please bear with me um so uh welcome all of you to this session on sophisticated inference in IMDb so here we are going to attempt to model uh some sophisticated inference simulations especially the one in the original paper using the prime DP module and how it is not part of time DP right now but we are in the process of adding sophisticated inference to the Pym DP module and I'm going to mainly talk about the code that I have developed to kind of add that functionality right so I'm Ashwin Paul I am a finally a PhD candidate uh at Monash University and mostly I work with active inference models and try to understand uh how to use them uh as an explainable uh model too basically understand emergence of intelligent Behavior right so let's dive into the material uh that I do it so to give an intro of free energy principle I'm sure all of you right now have an understanding of what it is but the central idea is that an agent is always trying to minimize uh the entropy of its observations right so if an observation is having really low probability uh in in your mind and and that happens then you are probably surprised and vice versa right so and uh the entropy here is defined as the information theoretic entropy where if you have a low probability then automatically this is a high surprise or a high info pick observation and as we all know active inference also gives us a methodology to Define what we call an agent environment Loop and this lets us Define what is the agent that we are looking at and what is the Behavior Uh of the agent given the environment around it and so on right so you're also familiar with the idea of Marco blanket and this is important because we always has to remember I mean have to remember the difference between the generative process and the generative model which is quite a famous point of confusion in inactive inference literature and for the people who try to understand it in the beginning right uh so the central question is that how does an agent minimize entropy because how does an agent know which observation is low or high probabilistic right so that is by maintaining a generative model and uh the

generative model will tell you which is a high probabilistic observation and which is a low probabilistic observation so the idea is that all the agent has access to is an observation that is coming from a generative process which the agent cannot directly observe and uh an intelligent agent will try to build up a generative model in its mind which is a model of the Hidden States and the observation it has access to and it can hope to kind of compute probabilities using this generative model right but there is a problem that in general it is an intractable problem uh to kind of marginalize the probability of observations from the generative model and that is why we have to Define an upper bound on the surprise that the agent is trying to minimize and there comes the idea of free energy right so this upper bound the agent is supposedly minimizing is the free energy and that's why it is the free energy principle and here we have a new term called Capital Qs which can be interpreted as the belief that the agent is maintaining about the hidden steps in its generated model and this quantity is the free energy and traditionally we see uh this variational free energy being interpreted as uh in in mostly the machine learning way where it is a balance between complexity and accuracy of the model so when minimizing the free energy the agent is trying to come up with a single model but at the same time an accurate model because here it's a minus sign with accuracy right it can also be interpreted in the physics way where the agent is always trying to minimize the energy of the model but at the same time maximizing the entropy of the model which goes in conjunction with the maximum entropy principle and so on from the classic literature so given that this idea of a generative model is so important um in in a software point of view uh that's the first thing that you might want to do right to define a generative model for the agent which is informed or not informed depending upon uh the experiment that you're trying to model So In classical active inference usually decision making is defined in terms of policies so for example if you are an agent in this environment so in the Mario game uh Mario is the agent and everything else is the environment and Mario has um three available actions um run jump or stay in this environment and the classic definition of policy is that it is a sequence of actions in time so if you have a time Horizon of capital T then a policy is nothing but a series of actions you might take so run run run jump and so on in time so this is the superscript is the action so here it is Jump run and so on and the subscript is the time and then then what you can have is a policy space which is a collection of many such policies ah smaller π_i and what you essentially do to take decisions in active inferences uh compute uh not optimize compute the expected free energy of uh every policy in your policy space and basically that can be interpreted as a balance between risk and ambiguity and yeah so when you compute this

expected free energy what you're trying to do is minimizing uh the risk uh that is how different is your belief about the observation uh from your prior preferences Capital C so this is also part of the generative model when you're trying to model control and at the same time you are trying to minimize the ambiguity when you are choosing a policy that has the minimum expected free energy right but this formulation has a problem that a policy space quickly becomes intractable that there can be an enormous number of small π 's or policies in your policy space and sitting and Computing the expected free energy for all such policies even for a small time Horizon is not possible but uh this is the classical structure that has been implemented in Pym DP uh nonetheless where we have different modules in pi mdp that is meant to be uh implementing different aspects of behavior so for example uh for inference or perception we have belief propagation fixed Point iteration uh marginal message passing and all that implemented in the inference module in the control module we have a different methods to evaluate expected free energy for policies one depending upon the expected utility uh the other one depending upon the classic method that I just explained then we have a module for learning so we learn the parameters in the pom DP like capital a capital B so likelihood transition Dynamics and so on and then we have algorithms for implementing all this in the algorithms module and then the most powerful thing in pi mdp right now is the Asian class where it is easy for you to kind of Define the agent environment Loop and we are trying to build up uh so today I'm going to talk about an Asian class that implements sophisticated inference um rather than the classical active inference um that we just saw um so as I mentioned um how many valid policies can be defined say for a Time Horizon of 15 in classical active experience right um so the first policy of course uh is a series of first action that is jump then you can have the last Last Action changed and and you already can see that uh there can be n number of combinations and for this simple case uh the policy space is as big as 10 to the power 13 and in a stochastic problem setting uh there is no way to kind of come up with a small subset of this policy space so that you can tackle this problem of computational complexity and yeah as I mentioned in a stochastic problem setting it is an interactable size policy space uh there comes the idea of sophisticated inference where we are thinking about taking this instance in a different way right so rather than thinking about sequence of actions in time we can directly think of what to do when we see something depending upon our beliefs about the current state and beliefs about the future right so if I see myself in a current situation what should I do and that's more like a straightforward thinking of how to take actions and here we expected free energy the structure of the expected free

energy is the same but we are not evaluating expected free energy of uh policies but uh expected free energy of uh observation action combinations so if I see something and if I do this what's the expected free energy and that's what I'm trying to minimize uh that's what I'm trying to uh optimize uh in in this setting right so here again we have the risk term where we are trying to minimize the deviation between the belief and the prior preferences we also have the ambiguity term uh and this together makes up the expected free energy of uh this time point at time T but we also have an expectation about what's the expected free energy in the next time step and um to evaluate the expected free energy of next time step you will have to again compute this equation with OT plus two and and for that you will have to again compute this equation uh with OT plus three and so on and this automatically becomes a research because of the recursive way this equation is defined and it comes with its own problems where there are clever ways to get around them and that's we are going to discuss that in the code today so given this structure of sophisticated inference here as I mentioned the research replace the policy space that we saw for the additional active inference we are used so in this Workshop what I'm focusing on is how to kind of Define a generative model and and given an environment so for example this is the grid that is stimulated uh simulated in the original paper and we are going to talk about how to build up a generative model for this grid that can be used in the sophisticated inference in in the pi mdp module and so basically what I'm trying to talk about is that the environment will have a step function that takes an action from the agent in biomdb and the agent will get an observation out of that action and we are we'll talk about this particular function agent dot step and agent dot steps will step will take up an observation and try to give come up with an action for the next time step right and this creates a loop and cleverly designing this Loop will let you see emergence of purposeful behavior in the sophisticated imprint setting so for example in this particular grid with sufficient planning Horizon you will be able to see that the agent is capable of navigating in this grid and so on so this is the example that I am going to focus on in this laptop today I want to excuse me I want to right away jump in um to the code so we have the pi mdp home which I am I'm hoping you are familiar with so we have this GitHub repository where we have the pi mdp module and inside time DP module we have several uh parts for it so here we have the agent in the original Prime DP module that implements uh the so-called classical active inference we have several environments and we have helper functions like learning in current smacks and so on so this is the module of time DP but in the parent folder we also have examples where there is tutorials about how to use the agent class how to kind of deal with the environments and so on so if you

look at the pull requests so we are right now trying to merge sophisticated inference into the original Prime DP module and today I'm going to talk about the code in this school request so if you are if you want to try this uh Hands-On you might want to go to this page where this full request is there and it has the same structure of time DB it's basically designed using by NDP and here what we have additionally is an agent SI which is a sophisticated Enterprise agent which does everything and also planning and decision making in the sophisticated inference way and in the parent folder there is also an example folder uh for sophisticated inference demo and what I am going to do today is to walk you through this tutorial of sophisticated inference and on the way I'm going to discuss and at points uh where I reference the helper codes I am going to go to that code and try to explain what's actually happening and how we complete that age in the environment Loop where we can see that purposeful Behavior right so yeah so that's the prime DP home then I also talked about the pull request so let's right away go into the jupyter notebook so this is my local copy of this repository so it's easier for me to run it and show it in my personal computer so this is the parent folder with Pym DP and examples and in inside examples I have a demo holder for sophisticated inference and this is The Notebook I'm talking about right so here in this example what we are trying to do um deal with this particular gridual task from the original sophisticated inference paper and make this agent or enable this agent to navigate to this red dot which is supposedly the gold state in this particular task given our prior preference like this right so this prior preference is quite informative in the sense that we write we can see that this is the most preferred State the white color and the surrounding states are kind of less preferred but more preferred than the far away ones right so this is the grid World task that we are trying to use so the first cell is importing all the necessary libraries and some useful libraries like numpy and matplotlib and the most important one is prime DB so I'm I'm actually now calling the local copy of my Prime DP with the sophisticated inference implementation not the original one which is not merged yet and the first thing I want to talk about is the environment itself right so the environment dot step part where if I get some action how does the environment work and for that inside this folder I have and I have a file which is great environment uh SI Dot py and this is basically an environment class so don't worry about how this environment is actually implemented the only thing to worry about is this function that we are going to use which is environment dot step so this function will take an action into it and depending upon the current state of the environment it will calculate what is the most probable next state given this action from the agent so that's the idea and then it will also calculate the reward of some a negligible

negative value if it is not the goal State and if it is the gold state it will give a reward of 10 and that's how the environment is designed and it will update the current state to the new state and basically what it will return is a new state depending upon your action um the reward for that action and whether it is an end of the episode and so on so this implementation is the standard open AI environment implementation and this is the environment.step function right so in this grid for example if I am right now in this state and if I take an action up so I have four available actions um north south east and west so if I go north then the environment dot step will make sure that I am in the state that is about the state and if I go east or west then I'll stay here or South I'll stay here so that's the idea and yeah and there is an episode length limit that is eight here that means that I am restricting the length of every episode to be eight which is the ideal length of reaching this gold state just to avoid confusion so in this environment after eight actions the environment will terminate and if you have to kind of Reach This rule state in the optimal time point so that's the idea how the environment is implemented okay uh I hope that is clear and there are many helpful functions in this environment like rendering the environment rendering a pride preference Matrix in this environment so if you design a prioritical friends this environment can show that in in a pictorial way how your prior preferences that you will see in the notebook below so now it's time that we Define a generative model for the sophisticated inference agent right before that let's define the structure of the generative model that we want the agent to have in its mind which is tailor made for this particular environment right so here in this particular grade World task we have 25 valid States um starting from this state all this black States in this path are valid States so there are 25 valid States and then there are four available actions for the agent north south east and west so this is part of our generative model this is also in alignment with the reality of the grid but this is about what the agent it has in its mind right and then the observation is just the state space the agent is uh it the problem is fully observed so there is no ambiguity there then we define basically the number of states then we define basically the number of states which is a list of your state space number of factors which is now one because you only have one heat and state actor here then number of controls which which is going to be four uh that's for available actions and your observation space like that so this is the structure of your generative model and let's look at the structure of the parameters now inside a pomdb so the first one is the likelihood function which is often denoted by capital A and here it is a function of how many observation modalities I have and how many state modalities I have right so if I run this cell yeah I have to run the parent cells to make sure everything works so I've rendered the

environment the structure of the generative model and here I have the capital A Matrix which has a structure 25 25 that means that I have 25 States and 25 of surveys and here because it's fully observable I am initializing it as an identity Matrix of size 25 so that's my likelihood Matrix that I'm initializing for this particular gradual task then the second element is the transition Matrix so there so please note that I'm using all the existing time DB functionalities to define a random a matrix and then using an identity Matrix uh on top of that so yeah I'm not doing anything new here it is the existing time DP functionality then what I can do now is Define uh the B Matrix which is also called transition Matrix so the transition Matrix encodes transitions like where I'm going to end up in the future if I start from a particular State and take an action so that's the idea where it depends upon the number of states which is the hidden State modality and number of controls so it has the structure of State action state where if I take an action from a particular state where I'm going to end up so that is also a future state right I'm going to initialize it as the true environment state so now this is part of the environment that I've built it will give out the b Matrix um it might be worth it to look at the structure of this B Matrix so here um we have 25 25 4 so that means that if I take an action from a particular state where I'm going to end up and we have the true transition Dynamics for this particular grid by Design so there is a function called get ruby and that will give us the true B of the system which can which the agent can use okay so ideally we would want the agent to learn this but for the purpose of this demo we are assuming that the agent already knows the structure and then comes the prior preference which is uh interesting here in the sense that it is defined as how closer you are to the goal state so if you are at the gold state then uh clearly that is the most sought out State you prefer that the most and how how do you prefer the neighboring states right so that is dependent upon the square root of the distance or basically the distance from that particular goal state so you define a grid which is 8 cross 8 the same size as this particular gridual task and then we have a method to kind of add values which is the preference you have for every state and if you render the particular C Matrix you can see the structure which is the same where this gold state is more preferred and the surrounding states less prepared and so on so now we have the C Matrix also defined in the classic Prime DP way and then I initialize that c Matrix as the C Matrix we evaluated in the previous cell which is small C this particular C Matrix okay and then lastly for the generative model we have capital D which is your prior over hidden States and for that I'm using a uniform object array so that means that I don't have a prior of where I am starting so let me run the pending cells so here the D Matrix is a uniform distribution over hidden steps I don't

know where I'm going to start the simulations and form so this is the basic structure of the generative model and then we have the agent class which I want to discuss separately like the environment right so given these environment parameters how would you expect the agent class to work so where is the agent class inside this folder structure inside the pi mdp module folder we have an agent SI dot p y which is basically a class again and similar to the environment class here also we have a step function where this will take an observation in through the function and also a flag whether or not to learn the environment which which is optional so if you disable it it will it won't learn the generative model if you enable it it will update the parameters of the generative model and what basically it does is it will return an action which is to be taken at this point of time and the environment can basically use that action right so in this file we have the agent class which I will explain in detail so I am basically importing that agent class in this cell and then we are going to try and reproduce this behavioral result from The Originals of State influence paper okay and for that so what we expect is that given this prior preference structure there are local Maximas in this prior performance structure so if you start from this particular point if you do not plan deep enough what you will end up is in one of these local Maximas where you don't see that there is a highly preferred observation say four steps down the line so if you are in this particular State what you will see is this local Maxima and you will go and sit there because the neighboring states are less preferred and this state which is more preferred is not accessible because of the world or the wall structure so you have to take a turn and pass through less preferred States and you need deep planning in order to enable the agent to do that the agent should be able to kind of simulate um four time steps ahead in time to see that there is this uh highly rewarding observation coming to kind of do that actions so that's the point that we are trying to um see in this particular demo so for uh low planning depth it will basically get stuck in one of the local Maximas but with sufficient planning depth it will navigate to the gold state so that's what we are trying to see right so we have different planning Horizons and what we are basically doing is give the agent a generative model which we right now defined the a matrix B Matrix C Matrix V Matrix then we have the planning Horizon of capital N so here I am um iterating over planning depth so n will be 1 3 and 4 for the loop then we have action Precision which is often denoted by Alpha in active interest literature so that determines which action is to be taken so a highly precise action procession means that it will stick to the action with the lowest um expected free energy but a lower accent action Precision is kind of probabilistic where it will also consider other actions then we have a planning Precision which is part of the

planning function we will discuss which is often denoted in the literature as γ then we also have a search threshold which is extremely important for sophisticated inference because as we saw sophisticated the inferences at research and research is bad in the sense that it can have a lot of computations but you have to Define a threshold to kind of ignore many possibilities to make it work and that's the idea that we will also discuss okay so Desktop review before we go to the class what we are trying to do is in a loop we are going to call the agent dot step and environment dot step in series so the agent will see an observation it will take an action that action will go into the environment the environment will give it back new observations and this Loop continues and we want to see over time how this Loop evolves into a purposeful behavior and and if the agent at all is capable for that so before I reveal the results let's discuss the agent class so in order to give an action when an observation is given the agent should have the planning and so on right so here is the agent the sophisticated inference agent there we are actually using the existing time DB agent for some functionalities so in Pym DP we already have a really well written functions um for perception um and learning so the only thing we want to kind of replace is how the agent is doing planning and how the agent is taking decisions uh over policies right so here here we are using that parent agent class so from time dp.agent we are importing that agent class which is sitting next to the SI agent that we are discussing now basically we are taking in the generative model structure from the main program for this class to work which is a b c and d and all the precisions and threshold parameter I mentioned then it is kind of um normalizing the prior preference that we mentioned in the main program so here if I look at how C is the structure of C is defined in terms of numbers and the prior preference is often interpreted or it should be a probabilistic distribution for the computations to work right so um we are going to normalize it as a probability distribution rather than um having uh having numbers that don't add up to one so that's what's Happening Here we are using softmax to do that then what we are doing is we are initializing um the existing time DP agent with these generative model parameters and what we are intending to do is write a planning function for a given planning Horizon and a given um threshold for three sets okay so um there are three functions in this stage in class one is a helper functions for planning which we will discuss now then there is a planning function itself which is going to do planning using research and then because it is a recursive research we are going to need a additional function that implements that recursive evaluation where we are going to call this function called forward search inside the function itself so we are calling this function inside this function so that is to calculate expected free energy for the next step

and it will call it again for the next step till we till our planning Horizon so that's the idea of recursive looping and finally it will return the expected free energy for all actions uh given and observations and then we just implement the step function where it is written sequentially what to do give an announcement okay so going back to the demo here we have this first idea where you get an observation and it gives out an action so let's go to the agent.step function and imagine what happens so if it is uh time t equal to 0 or in the beginning of the experiment uh what it is ideally supposed to do in the first place is in for that state using its observation so what we are giving it is an observation and using the modules for inference it's going to come up with a belief Q_s which is an which is a belief about States okay so self dot Q_s is the belief in inside the agent and once it has a belief about where it is right now it can Implement plan dot research which is um do planning for this particular belief of hidden state right now and once it has done planning it can take decision using the sample action function in time DB and basically return it uh return that action and if it and for every other time steps the sequence Remains the Same but it is also learning about the structure if you enable learning in in Eurasian class so that's the step function but in order to do planning what it does is it kind of reorganizes the generative model structure for any number of heat and state modalities and any number of observation modalities so to discuss how melting works I would like to talk about um the new A matrices and V matrices it it evaluates for implementing that planning and let's understand that so let's go back to the original hidden State factors so here we only have one hidden State Factor um so that's why it is d_0 right and B_1 does not exist because we only have one hidden State Factor but imagine that if I have two hidden State actors with the same size maybe um so here it it could be a location and maybe something else inside the agent's mind and we should also have controls for these two hidden State factors just like so there should be control for every hidden State Factor if you're familiar with active inference and then maybe the observation space is also directly observing these two date and state factors okay so this is a new generative model structure with multiple hidden States and multiple observation modalities and right away you can see that the dimensionalities of your parameters change you have 25 observations coming from 25 times 25 hidden States and if you look at the structure of the first observation modality it's the same but what we want is a new Matrix where it's going to be 25 with not two hidden States but just one written State it's only a reorganization of um the generative model but computations essentially Remains the Same that's what this helper function is trying to do which will make things easier for us when we have multiple hidden state modalities so um if you have multiple hidden state modalities we are going to

compute how many total states you have which is the multiplication of number of written States in each modality okay so if you have 25 hidden States in one modality and 25 hidden states in the other modality you're going to have 625 total number of scales and if you have four actions each in each modalities then you have total of 16 actions which is nothing but the combination of these four acts each in in in the modalities So you you're going to have 4 times 4 16 actions if you have two modalities and it's basically um going to build a generative model that is the same um model parameters but just with the different dimension structure so that it's easier for us to calculate the expected so now we have a new a and new B and a new belief which is nothing but tensor products of existing parameters and beliefs it's nothing but a new big Matrix and nothing else okay it's not a transfer it's not a change it's just a transformation of structure and given this a b and Q we are going to predict what's going to happen in the future and evaluate the expected energies for them in order to do planning so that's the second function what we are going to do is first call the first function which will do the melting for us and set up the generative model in in good Dimension cc to compute then we have the expected free energy itself for all the and then we have the probability that depends upon this expected free energy for these actions so why is it just the actions and not the observations because here we are going to evaluate expected free energy of actions uh for the given observations right so let me um go back to the slides and discuss this pictorially to make things easier so here we have the grid and we have the fire Footprints and what we are trying to implement is that if You observe some observation at time T then you are going to consider the consequences of your actions given that observation because you can predict what's going to happen because in your generative model you have the transition dynamics that will tell you given this state if I take this action where I'm going to end up right so that's basically predicting what's going to happen in the future and you're right now considering the consequence of um available actions in Your Arsenal and then if you take an action then you can predict what's going to happen in the next time step as a new observation right so you have a probability distribution that tells you that say this observation is the most likely uh and the other observations are not really likely uh then what you will do is you do this again you consider the consequence of doing your actions from that particular observation and and this goes on in your planning depth right so this can be thought of as maybe you want to go to the gym then you are going to consider all the consequences what will happen if I wear my shoes if I don't wear my shoes uh if I go in my car if I don't go in my car then you realize that okay I have to wear my shoes then you consider the consequence that I am now ready to go to the

gym and me going to the gym will end up me being in the gym so that's the idea like you consider the consequence of where you are right now and you can go as much as you want right you can predict so in a game of chess you might be in a particular State you consider your consequences you see the future you consider consequences from that future and you can go as deep as you want depending upon your computational abilities right so that's what you're trying to implement in this agent class where we are considering consequences yeah for every modality we are going to consider the expected free energy for this will basically call the next function which is forward search so forward search is implementing the thing that I just mentioned um considering consequences and in forward search what you are basically doing is for every action so in line 149 I have a loop that goes over every then I am going to consider the posterior or the consequences of all those actions I use my transition probabilities to evaluate that consequences then I am going to predict the observations because my prior preferences are defined in terms of observations I am going to predict my observations and then evaluate the expected free energy which is the sum of risk and ambiguity okay I hope that makes sense like here you have considered the consider the consequence which is consequence of future which is post or posterior and you're basically evaluating how good that posterior is depending upon your expected rate and that becomes the expected free energy for that particular action and you do this for all the okay and why this is powerful it is because you can go as deep as you want so here in the next step you go to this loop where you will check if I am crossing my deep planning or the depth of planning and then you are doing basically the same given that posterior what is the consequence of the actions of that particular posterior so here for considering that we are again calling um the parent function so the same function forward search to consider consequences of those combinations and it will basically come back and add up to your expected free energy so what happens over this sequence is that you consider some or all future consequences and then all that values will trickle up your tree and your that sum of the expected free energy will tell you which action is good or which action is bad which you can take um in to kind of see your preferred observations so that's the idea of implementing research and I will also talk about the importance of uh this threshold here uh which is which is which makes this algorithm possible so without this threshold this algorithm will not work I will explicitly talk about why that is the case and then once you evaluate the expected free energy for all available actions given the present State you can basically compute um what you call the action distribution that is how probable is my action or how I should take my action so that is and we also have this action Precision parameter

Alpha so if Alpha is very high then it basically is a highly skewed distribution where you will always choose the action that minimizes expected free energy if Alpha is really low then it's going to be a more um sparse or spread out distribution and then you can use this action distribution to sample actions and so in the Asian environment Loop we just finished doing planning and Computing that action distribution then using that action distribution you can sample an action from your policy space so let's look at the policy space in this generative so I'm switching back to the original generative model with one hidden State Factor let's do planning and maybe um initialize this agent I just want to initialize this agent to see the policy space not randomly I initialize this agent say for planning depth of one and if I look at agent Dot I can say that I have basically four available actions uh which is not Southeast and West and if I have an action distribution uh it will tell me how probable is to take that action so if I look at π so okay so this is not defined because I have not done planning uh but I can do planning and then it will be defined yeah so I implemented planning with research and then now I have an action distribution so for this particular scenario I am going to take my third action the most which is 0.99 basically that's the probability which is north south and east in this particular region I just wanted to kind of familiarize you with the matrices but we are now going to see um the agent environment Loop in action yeah so now you can sample an action from the sample action function and then Implement learning which is using the standard π mdp way where I will update my transition Dynamics and likelihood Dynamics depending upon what I see and what's my belief and so on so my emphasis is on the decision making part so once you sample an action then that action basically goes back to the environment okay so now let us implement this for a planning depth of one and see how the agent behaves so here if it is a planning depth of one then that means that the agent is only considering the consequence of one time Step Ahead just seeing the immediate future for doing planning right so I'm giving the planning depth of one I'm resetting the environment where the agent is going to start from that initial start state and in the loop it's going to get that observation take that action uh give back an action and we will look at the action probabilities uh and also we will give that action back to the environment get back the observation and this Loop continues till the episode is terminated and I have set the episode length to be eight just to see the outcome of a tax so when we run this Loop these matrices are nothing but the action distributions of How likely each action is to be taken and this is where the agent end up in the last text okay so let's maybe kind of also enable this `environment.render` that will show us where the agent is at every time step so initially the agent was at this location and we have an action distribution

of this here so north south east and west so the agent knows that it should go north why because if I look at the prior preference um this state is more preferred than this state right so the agent successfully calculated the expected free energy and inferred that okay I should go to this state and not stay in the state and because it has the generative model of transitions available it can infer that I should take that action North to go to the so that's good and the agent goes to north and at this particular State the agent infers that it should go to north south east so it will take an action East uh and it will go here and at this point of time uh I want your attention where the action distribution is equally probable for north and east why is that because the agent is only looking at the immediate future right so let's go back to the prior preference where the agent is right now here or is it here yeah it is right now here in this particular State and if the agent is considering immediate consequences of just one action then these two states are equally good for it to be in the next step so there is no distinction between these two states if it is only looking at the immediate future so that means that the expected free energies will conclude that I should go to North or east it doesn't matter if I'm looking at just one time Step Ahead okay that's the idea and out of probability it is going here it took the action East and from this State when it's doing inference it's inferring that this state is better and basically it's end up ending up in this local Maxima state which is this particular state where the neighboring states are less prepared and this is wall and you can you can't go there because it is forbidden for the agent by structure so it's basically going to sit there forever where it only sees that local Maxima of priorities let's look at what might happen if you have higher planning depth so if I go to the planning depth of 3 um then that means that the agent is actually reaching the gold state at the last time step but still in the third time Point uh it had two probabilistic actions not an East so here from this particular State out of probability it took the action North but it cannot equally take the action East and end up in this local Maxima so let's run again and probably it will end up in this local maximum okay and only for the planning depth of 4 which is sufficient enough uh which is necessary for this particular grid um the agent is fully sure of what to do so at every time point it is fully sure of what to do uh that it has to go north then east north north east east and south and reach this particular state so only for time step um or planning depth n equal to 4 it can successfully navigate this grid um it's 100 certainty so that's the idea next implementation that I hope you got so there's also this idea of action Precision so here it's a high action Precision that is why it is taking the actions that is from the probability if it is a low action Precision um like one then the good actions are more probable but that doesn't mean that it will be taken right here by luck it is taking the right

actions and reaching the state but here probabilities are most passed but you will also see like exploration Behavior Uh in more number of Trials if you control this action process so I set it to a high value to make sure that the agent reaches the goal um for this particular problem but it's worth playing and it's important yeah so for different planning um depths like one three and four in this problem this is the behavior that you expect for lower planning depths which is not sufficient the agent ends up in local Maximus and or local minimas of expected free energy or local Maximas of fried blueprints but with sufficient planning that it's able to navigate and reach the goal so that brings us to the last point in this tutorial where why is it important to have a threshold in evaluating substituted inference so by threshold what we mean is that we can ignore future possibilities in two levels right you can ignore um not likely actions or not likely observations in this research but if you consider the consequences of all actions and observation that means that you'll have to consider four consequences in the first place then you will have to consider four times the action states in the next time step for the next time step then all of that multiplied with the number of actions and this research becomes intractable until explode and it's even worse than the classical active inference policy space problem but by defining a threshold of even a small value will ignore possibilities so where is that implemented in the forward search algorithm um we are considering actions uh with only action probabilities greater than the particular threshold here I am defining it as 1 by 16. uh also in the parent paper it's 1 by 16. um the action probability okay uh so if it is 0 then that means that it will consider all the consequences of future and that's interactable so you can ignore actions which is not probable and also ignore states which is less likely or only consider States um that has probability greater than this particular threshold value and that significantly reduces the computational complexity where you will only consider combinations that are probably in in the future uh tree and that lets you go deeper in your planning Horizon that's an important Point here and if you compare the time that takes for deeper plannings uh for a search threshold of zero uh so address search threshold of 0 0 means that you will consider all consequences and the more deep you plan um the more time it takes and if you see uh to consider only the first future or the immediate future it takes 0.01 seconds for considering three possibilities into the future it takes three seconds and for four it takes 300 seconds then you can see that the computational time is exponentially growing but if you have a very small search threshold um you have that computational time that makes sense for implementation uh in in real world right so here for n equal to 4 that is four time steps into the future it's only taking 0.1 seconds and that's okay it I can still do simulations with this complexity but there is

no way I can talk about how this is um how less the computational complexities it truly depends on uh the nature of your prior preferences and environment in action but this such threshold actually works in in real life and we just saw that in our simulations right for n equal to 4 it took only like 0.3 seconds in our simulations but if you set the search threshold as zero um it already is 300 seconds for doing full debt planning and if I set uh planning depth of five then it will basically Run Forever maybe again I I would not be able to do symbols so that's the idea of search threshold um so actually that's it I wanted to explain the age and class the environment class and the particular demo and yeah maybe it's a good time for questions if if anybody was listening uh and I hope people um get to play with this code um and look at the tutorial and and implement this and build generative models like this so this particular example is how do you build a generative model for this gridworld task and see how the agent is able to take meaningful actions but here we gave it the true structure of the environment in the B Matrix and a matrix and so on but you can also play around with learning in in the sense that while defining the agent dot step you can add a flag that says learning equal to true and if you start with an uninformed a b and so on you can experiment on how the agent is learning that environment here you can look at the B Matrix in the beginning you can look at the B Matrix after say 10 trials and see how the learning is taking place here it doesn't matter because the agent knows the structure and it won't learn much but if it starts from unknown structure then there is scope of learning also to be implemented and it's already implemented because we are using uh existing Pi mdp functionalities uh for learning A and B and it's already part of the step function so I hope step function is clear um which is the only thing you need to know if you are trying to implement sophisticated inference and just the names of these matrices if you want to probe them and look at them and yeah I hope the session was useful so I thank my collaborators and Corner who maintains IMDb and also Brendan who runs the prime DP Fellowship which I was part of and that's where I worked on implementing sophisticated interns in my mdp and it will be part of the original Pi mdp module soon and I hope people can start using this module to simulate so stated inference experiments and this basically becomes so maybe it's a good time to discuss questions or clarifications on the code or maybe it's a good time to take a break yes thank you that was awesome well I have a few different questions and I'll read a few from the live chat so I'll first just go to the live chat and then um ask those and then ask some other questions so Dave asks how do you think about the neural implementation of recursion brains don't seem to implement computer hardware style recursion deeper than a stack depth of one aside from heavily overlearned tasks we can

confine ourselves to asking about recursion for the purposes of exploring temperately deep State spaces searching forward in so how do we how do we reconcile this really beautiful and elegant and computationally efficient full depth research with the biological basis of multi-scale planning um so I'm not an expert in neural computation but the answer to that would basically you're doing only one competition at a time right and your have or all you have need is some memory to store your beliefs and use those beliefs to kind of do the same computation again so um we are not talking about uh neural I mean this hardcore recursive implementation we are only doing local computations at a time and just because of the structure of the generative model and we and because we have memory um this this can be done and and I don't see why and brain can't do it even though individual neurons might not be able to do it the main brain has memory the brain has the ability to store memory and the ability to dream the ability to simulate uh it knows the consequences of actions and you do this on a daily basis where um you plan your future and decide what to do right so on a single neuron level I'm not really sure of how to answer that question but I don't really see why the brain can't do it as an organism cool um okay to the code I guess I have a few questions let's can we go back to the maze and of course if anyone else has questions in the live chat just go for it so in in the Maze how how does the uh the moves that are possible how is that reflected what what stage is it updated that what for example that it initially knows that it can only go up and then it can only go right or down where is that reflected with the updating of what what the like kind of that relational aspect of affordances what is even possible to do and then how does it evaluate in a deep tree search when it does does it need to know what things could or couldn't happen in the future um so if I understand the questions correctly there is import there is an important distinction between the generative process and the generative model right so in the great world um which is the generative process um we have implemented all of that manually where we have this transition Dynamics um that already knows what will happen if you take an action from a state so it's like an environment that knows how to work um so it's like a reality where there are consequences of actions and it's already there but this information is available to the agent uh to be part of its generative model and in the generative model what it basically does is use that transition Dynamics which is given or learned to simulate what might happen in the future okay so once I have that transition Dynamics um if I go to the if I go to that research what it is essentially doing is evaluating the posterior given a particular State and an action J so from my generative model I know that if I start from this state and if I take this action I go to this post areas and I consider all the consequences and if in the generative model it is

unlikely that if I go east I don't go there and so on it's basically it's it's automatically reflected in the expected um so I hope that answers the question um if I set the action Precision to be high also enable the so initially I am in this particular State and we with respect to the expected free energy what I'm doing is I'm using my generative model to predict what will happen if I take four actions and my predictions will say that if I take North I will go here and if I take all the other actions I stay here and because in my prior preference North is more preferred I can infer that not just uh action I should go for so that's the so here the grid structure is given to the agent and it might be a little confusing but the agent can also learn the script structure and if this will work so once I know the grid structure as an agent I can simulate the future and consider the consequences of action so that's what's happening and once I'm in this state I will consider the consequence of all four actions and I will infer that okay um going east is better because that will take me to this state so there's always a difference between or you should keep in mind how generative model and generative process are two different things and what the agent knows might not be the reality or might be the reality depending upon what you give it if you were gonna go about making a new situation that you wanted to do generative modeling for do you tend to start from an existing working Pi mdp notebook and start modifying State spaces or do you draw it out on a canvas like how would you recommend somebody other than replicating what is shown here let's just say we're interested in something that's not exactly just a maze what do we do to get our head around how we should proceed good question so if you're trying to simulate say a new environment you have the heavy lifting to do which is to define the generative model for the um you can either Define a various parts generative model which the agent can learn but you have to define the structure that structure should be there and only using the structure the agent uh can learn the generative model so here um you can make use of this cell of code to understand how I Define the structure of the grid okay so I am defining a structure for the agent for it to make use of I'm saying that there are 25 valid States and there are four available actions and this is the standard way of defining the state space in time DB and you also have to Define um the Central parameters a b c and d for the agent stimulations so here I am um a using my state space and observation space but this step of giving it or telling it that it's an identity Matrix is my decision choice in my modeling I don't have to do this for simulations I can see if the agent is learning it from a random a matrix or when it starts from a random Matrix and similarly for this B Matrix this structure is defined and there are functions like this which can give you a random B Matrix but this is a modeling choice where if I want to give it the grid structure or

not give it the grid structure I can start from a random V Matrix let the agent learn and look at that learned beam it just for the purpose of the demo I gave it the grid structure to enable it to take the actions but it's not a mandatory thing so this notebook is useful in the sense that you know what to do but definitely you should play around with steps that may not be mandatory right so if I give a prior preference for this state to be the maximum then you can see behavior that the agent will try and go there um this prior performance is defined uh in conjunction that this is the goal state but this may not be the cold State and in a different task what prior preference means is different according to the environment um right so that's also there and your prior so once you define that generative model which you have to do that's you can't run from it then everything is kind of automated the agent only have to use the agent dot step give it the observation from the environment the agent knows how to take actions and then everything that happens inside the agent you don't really have to worry um yeah so this structure I'm sure will be useful for your particular task that you are trying to model in your hmm yes very interesting and how would you contrast that or or point to any similarities or differences with how this would be pursued outside of active inference like if somebody were just going to use another kind of deep policy agent in the Maze example um what parts of the process would be familiar and what parts would be like a lot of work that wasn't expected or skipping through parts that were a lot of work otherwise um so the general structure is very familiar to somebody who does things like this in reinforcement learning so um the idea that it's an agent dot step and environment dot step so this is the standard open AI Jim way of writing an environment and a standard open AI way of writing an agent okay so if you have a queue learning agent who does the same trying to navigate then the way you have to define the Q Matrix is the heavy lifting there it's just a state action mapping and in contrast to that in active inference you have to come up with a generative model that you want to see so in active inference it generative model is the central thing right without that without a generative model there is no meaning of purposeful behavior in active Independence so the only unfamiliar part for a person who's coming from uh say a field like reinforcement learning is the structure of the generating model but there is no way other than getting used to it where it's the form DP structure which dominates but if you're doing deep active inference all of this is going to be neural networks and palm DPS are also not active in prints um things right it's a industrial engineering thing so form DPS must also be familiar for people who are coming from the computer science background just the idea that um what really happens in the agent is the active inference part where we have expected free energies and variational free energies and if

you want to learn about that then you have to go to the agents and see how it works look at the matrices numerically see what's happening but I don't in in a level where you want to get started I don't see any problem all of this is standard Frameworks like bomb DBS and openai gym environments agent environment Loop all this is very deeply discussed in computer science it's not um so what what other um motifs or cognitive phenomena are you excited or how do you see the pi mdp development trajectory continuing after you're sophisticated inference gets pulled in so the IMDb had the original functionality and um the functionality to implement or simulate um General active inference agents with the policy space and so on and that enabled a lot of people in the community who is not who are not are familiar with um complicated coding and so on who people who do psychology Psychiatry and and all the things right so whoever want to come up uh and try Implement active inference by mdp enabled that and I'm I'm hoping that this module will enable people who want to try out so stay great inference experiments in their particular domain so this if you spend some time and get familiarize yourself with the structure of how Pi mdp works then everything else is just writing a jupyter notebook with minimal code right to simulate this so if you have a particular task in your domain I don't see a problem for a beginner to kind of try and code it and what I am very excited to see is people using this module for variety of experiments just like how people started using pi mdp and sophisticated inference is taking off and it's now widely talked about about how it is how it is the way of doing active inference and I'm really hoping that people in various domains start using this module and see their experiments and I look forward yeah so what time DP did two years ago I'm hoping this module will do to people who are trying to uh model active inference in this state so you mentioned the open AI gym and the standardized format and and what benchmarks do you use or what are you what kind of test Suites are you comparing and how do we really know when we've made a generative model that really exceeds or excels in a way that other um techniques are just not doing yeah so if I um may go to the open AI gym website um we have several experiments there the classical reinforcement learning examples um like the lunar lander that you see in this screen right now so active inference permits Inception has faced problems of scaling um to tasks and that has that's in itself a field of research in active um scaling active in print and that's one of the reasons why deep active inference took over um dealing with tasks like this so there are benchmarks um even now where the sophisticated inference may not be able to deal with State spaces and personally that's my research in my PhD I am actually looking at um optimal optimizing computations in sophisticated inference algorithms that lets you scale up two

environments like that but to get started you will have to kind of write code and see if it works for an environment then look at if it's not working then you have to look at method to scale it up and and so on so if I am talking about benchmarks sophisticated in Princess as good as any RL algorithms um for this state space so for small problems so which Gate entrance will work and it's really good but for high dimensional problems like this um the classical implementation that I just showed might not work but it's good enough for any decent experiment but if you want to scale up then that's still open and it's a new field of research and what you do might become a next new important paper right so that's all I can tell in that regard what measures do you think you'd be looking for like a computational resources or what what are the measures that even make sense to juxtapose such different methods yeah so the open AI gym was designed for that to compare different uh algorithms so open AI gym by definition is a collection of many environments so in my demo I was talking about the grid open aigm is nothing but a collection of many environments which will let you interact with those environments using the environment dot step function yeah so here we have the environment.step function that will let you interact with the lunar lander and that particular task will have matrixes that lets you um judge how good or bad your algorithm is so in this lunar lander problem how optimally can you land your Rover between these two flags by spending minimizing the fuel and so on so those matrices are very tasks specific and that's One Direction you have to take you can take uh try and compete with RL algorithms in in matrixes but um the right potential or the potential IC in sophisticated inferences uh modeling intelligent Behavior where in RL the focus is to get things done to make this work but it's not really explainable especially deep RL and deep learning methods but in active inference if you manage to scale it up they are explainable and and they are um that will let you understand how intelligent emerges with time and I I see that more interesting than um competing with RL because if your focus is getting things done then maybe engineering is the right way and not active insurance any other comments or thoughts Osman do you have any other comments or thoughts no I I'm pretty happy I hope um I was clear explaining the code maybe it was too complicated or simple depending upon your level um but I hope it is useful to at least one person who would start using this and write the code and so thank you so much for your time awesome thank you for the opportunity then thank you for joining till next time see thank you so much bye greetings to Korea until next time see you you can um mute the live stream or turn off the other live stream but yeah thank you for joining yes uh thank you for uh inviting thank you for it yeah um well we have a little bit 35 or 38 minutes so it would be awesome to

uh have your presentation right uh so the moment uh well can you see my well but I saw PowerPoint okay uh can you see my screen uh perfect perfect so shall we start yes thank you okay oh then uh thank you for uh organizing this wonderful uh Symposium today I'd like to talk about a relationship between canonical neural network and active inference and possible extension uh or modern social on the shared intelligence using canonical neural networks so let's start so uh as you know the free energy principle is proposed by a car free stand that states that perception learning and action of all biological organisms are determined to minimize the variation of energy as a tractable proxy for surprise minimization and by this uh process organisms can perform uh parational Beijing inference or external mineral and this is this cartoon just show one example typical setup under the free energy principle active inference here there is a hidden state in the external world and only a part of this state can be uh observable for our agent this dog and this transformation is done by a genetic model parameterized parameterized by Θ and to in for the hidden state agent need to reconstruct the copy of the external State called posterior relief and this optimization is done by minimizing a variational free energy and parameter is also optimized by minimizing free energy to to obtain up generating model to represent this relationship an interesting aspect of the free energy transplant active inference is its application to the optimization of action here our agents have some preference prior C and to obtain this observation in the future agent May select the action that minimize the expected free energy in the future to to to obtain the most predictable uh outcome and finally by selecting uh actions that minimize the expect frequency it can obtain the feed so this is a typical setup on the active inference but question is what is the neural noun substrate that can implement this process so that that is our interest and to address this issue we propose a salary as borrowed here we consider that uh external world is parameterized characterized by a set of the variables like this here this is a posterior expectation and as indicate the hidden state in the external World Δ indicates the action of the agent or decision and the Θ indicator parameter and the Λ is a hyper parameter so those uh set or so parameters or variables characterize a generative model and free energy variation of energy is defined as a function of the sequence of observation and the external state and its minimization indicates the variational free energy operational elevation inference uh similarly we consider a dynamics of neural network and we consider that a Dynamics is characterized by a set of those variables here x indicates that neural activity by having great at the middle area neurons and the Y indicate near activity of output radiant neurons and W here is a synaptic weight and the Φ is the other any other free parameter that characterize the

neural network and we consider that neural network Dynamics is characterized by minimization of some cost function here this is a function of the O and the Φ internal state of neural network and so its minimization indicates the generation of neural network Dynamics including both activity and plasticity and our Theory indicates equivalence between those two functions means that for any neural network that minimize some cost function error there is a generative model that satisfy if we call Air which means that uh uh the recapturing of external Dynamics is an inherent feature of any neural system that for such Dynamics so this is an interesting prediction about the two other struct so I I would like to introduce some analytically tractable example to understand uh this relationship formally so first we consider a very simple architecture this is represented by a pong DP without any stage transition so hidden State s is simply generated by prior distribution D and it's uh it is a binary state but we consider a vector of binary States so it is a factor of factorial structure and observation is also binary vector and the transformation from s to O is characterized by categorical distribution uh using a regular Matrix a and variational Beijing inference indicates the inversion of this generative process like this so by solving the appropriate free energy functional we obtain those uh posterior brief that is the optimal in the Bayesian sense so on the other hand for neural network we consider such structure uh here upper part indicates the generation of signal in the external world and the neural network comprises only a single layer here activity of output layer are generated by a weighted sum of sensory input or weighted by synaptic Matrix W and we then characterize this neural network in the next slide so uh to model neural neural network or neuron we start from considering project equation which is the four which comprises four differential equation and this is a very complicated non-linear equation and although it is laboratory possible so we consider some reduction of these equations so a typical reduction method is is like this for example M here is a much faster than other variables so this can be replaced with its fixed point or it is known that H and minus one sorry it is known that H and $1 - n$ is a have a similar Dynamics so we we can consider a new effective variable U to characterize those two variables and then we obtain to the positive equation like this so this is a famous class of neural network model which includes the well-known fixed View and agreement model uh or other uh continuous neural network model and we we modify those models to to derive canonical neural model so this is a definition of canonical neural network model here leak current is characterized by inbus sumoid function instead of a cubic function which is adopted in the fifth few nagma model and we also consider a connection uh synaptic connection and this part indicate the synaptic input from a sensory

layer through uh weight matrices and one may consider that W_1 indicate excitatory synapse and w_0 indicator hv3 synapse and the threshold adapts thresholds that are function of W_R and w_0 interestingly when we consider the fixed point of this differential equation we obtain a well-known red coding model in with the sigmoider activation function which means that we can say that in some sense this canonical neural network model is a approximation of how you can have three equation and its acts approximation level is a in some sense between the realistic model and the most simplified red coding model so we basically consider this type of neural network model and in the next slide we consider what is the plausible cost function or this neural network model so again we write the same equation for canonical neural network model which represents the activity of neurons and Vector of neurons and we consider cost function for this differential equation which can be obtained by simply calculating the integral of uh right hand side of this equation and get this type of cost function for neural network this is biological portable cost function in the sense that its derivative derived a neural network activity which has a certain virus called possibility moreover if we consider a derivative of this cost function with respect to synaptic weights w we obtain a conventional cyanotic plasticity rule which follows the hebian law uh on the other hand for Beijing inference we first Define generality model uh like the previous slide and we then derive the sorry we then derive a variational energy for uh given generative model so this variation of free energy is derived from the Palm DP model in the previous slide and its minimization indicates the page and infrast and running so we formed a formal correspondences between a component of those two cost functions like this so this block Vector formally correspond to this block Vector that represents posterior expectation and this logarithm correspond to this logarithm and actually this a matrix can be represented as the block Matrix like this and it's a DOT product uh correspond to this computation and finally uh these five uh naturally correspond to this uh log D logo of State prior so which means that uh because the cost function uh same it's derivative uh provide the the its derivative provides the the some sorry so because the cost function are formally equivalent it's derivative uh it's uh sorry it's a it's a result of uh derivative it also corresponding each other so which means that uh for any neural activity equation in this form uh there is a corresponding Bayesian inference equation this is a equation that computes the posterior div of the helium State and this synaptic process equation format correspond to learning or parameter or generative model and moreover by establishing this relationship we can consider the reverse engineering of the genetic model from empirical data here uh this uh schematic summarize our approach to reverse engineer generative model and we we first

record the neural activity and assign the canonical neural network to explain this obtained data and by Computing the integral we obtain the cost function for this canonical Network and by the mathematical equivalence we established we can automatically identify a genetic model and free operational free energy that correspond to this neural network architecture so interestingly this is a Bayesian agent which is a type of which is a kind of artificial intelligence but importantly this artificial intelligence is a formally derived from empirical data so we can say that this agent is a biometric integers that follow the pre-energy and then its derivative with respect to a parameter posterior derived synaptic plastic CR algorithm that follow the free energy minimization and it's time integral can predict the running process or original neural network data so which means that if if the friend as a principle is correct then this prediction should work should work and should be able to predict the the result of this neural data without referencing to the data itself so our strategy is uh in summary our strategy is that we reconstruct genetic model only from the initial data with like initial Data before learning and then predict the running process or running the curve that this neural neural system should follow by using the free energy principles and compare or examine where the this prediction is correct uh by by comparing uh capture data after running and the prediction by the frequency principle and if this prediction is correct then it indicates the predictive validity of the frequency principle and the setup considered so we apply this strategy to the in vitro neural network so here this in vitro neural network are stimulated uh using the form DP genitive process defined in the previous slide so there are two hidden sources that are binary signals and they are mixture and they are mixed to generate uh 32 sensory stimuli those are also binary and this is a overview experiment uh by by stimulating in vitural neurons so they generate Spike spiking response and those rhyme indicate a high density spiking response so we of that uh in some neuron the response specificity is so we found that uh for some neuron response was high to Source One uh signal compared to Source 2 signal and if we see the transition of uh those neurons we found that although we we remove the offset uh at the first session uh to to set those activities zero but we see that those neuron self-organized to responds High when Source one is one but not uh but those neurons response low low level when the source one is zero so which means that those neural activity through the neurons response uh was uh consistent with our theoretical prediction that new activity serve organized to encode the posterior expectation of helium State and we also found that some other group of neuron response preferentially to Source 2 but not to Source One so then we ask okay then we we found that posterior expectations encoded by neural activity and the next question what is

other what about other neuronal substrate and we then ask whether we they ask if the priority about hidden state is equal to effectively equal to the firing threshold of a neural activity model yeah Network model uh so if the server is correct this corresponds correspondence should exist to check this uh we we first simulated a Beijing agent and when we evaluate the priority of the Beijing agent uh the inference was attenuated as I expected and we warned that when we buried the excitatory level of in vitro neural network by using the pharmacological manipulation we also found that uh the attenuation the attenuation of so which is consistent with our theoretical prediction that firing threshold encode priority or with the hidden State and and next we consider whether the synergic plasticity followed the freenancy principle by asking whether frequency principle can predict the qualitative self-organization of a subsequent uh a subsequent neural data so here we model a neural neural neural network like this there are two Ensemble neurons that encode Source One and Source two and we first compute the effective synaptic weight of uh those uh networks uh using a conventional connection strings as mentioned approach and the plot dot uh final equate on the landscape or theoretically computed free energy so this is a trajectory of MP Curry estimated synapse weights effective synaptic process synaptic connectivity and as predicted uh those uh changes reduce the the free energy and here we computed this theoretically predicted free energy landscape only using the first 10 sessions data so that this indicated indicate some prediction of of the self-organization and uh for more explicit uh prediction we then simulated a new activity and plasticity using green as a principle we uh here the brighter color indicates the prediction of uh data without referring to activity data so these those lines uh exactly follow the this frequency gradient uh and we found that this predicted trajectory is a title title correlated with this empirically estimated effective synaptic weights and the earlier rate is uh practice so it indicates that in the principle can uh quantitatively predict the serial colonization in this setup so it indicates some predictability of different energy principle under this sector and then we also consider the modeling of the neural modulation uh using active inference it is well known that synaptic process is modified by various factors like dopamine or other learning assets recording serotonin so on and one interesting property of those modulations that as even so dopamine was added after associative plasticity was established it can change the result of plasticity in a post-hoc or retrospective manner so it's a it's in price some Association to the reward and plus decisions so we model this process using canonical neural network so here we model the uh post Hulk modulation of ABN plasticity using uh this type of uh plasticity equation and we also consider the recurrent neural network structure and output

create for this network and we consider that modulation occur at this connectivity uh a layer and then we found uh cost functions that can derive those those differential equations and then formed a corresponding variational for energy and GNT model so which means that this sort of a neural network activity including moderation of sinus plasticity exactly follow the free energy principle and some type of homedep generality model and by using this we show that this uh biological approachable neural network model with a moderation of heavy and plasticity can solve the some sort of uh delayed reward task like maze task and then finally we would like to discuss a possible extension to this framework to the modern social intelligence so uh to to infer our con specifics uh we we need to select an approach to create a generative model or or our partners depending on our partners so this can be done by vegan model Resurrection scheme and we a previously proposed a model that can predict the multiple valid forms using uh uh one big generative model that comprises multiple genetic moderates and this movie shows uh prediction of uh prediction of uh songs so like this uh this modern nicely identify which giant model is the best to explain a given sensory input and this process indicates the the and the the model can correctly input the appropriate model and then imitate the song by its own action although in the previous work we didn't discuss a detailed neuronal substrate for this mixture generative model we now be able to consider the uh corresponding architecture for example if we consider the modulation of module neural module by neuromodulator like dopamine it up add an attentional filter and this attentional filter can be explained by uh three Factor heavy and running rule introduced the introduced in the previous slide so again the this modulation is a worked as the post-hoc modulation no past heavier plasticity and this modulation can optimize each model to represent one generality model one General uh one song in a mutually mutually independent manner so through this process it is possible to learn multiple genetic model uh in a virtually plausible manner so in summary we found that the Dynamics of canonical neural networks that minimize the cost function can be read as a minimization of variation of her energy it indicates a frequency principle is an explanation for this type of neural network and we also validate this prediction using some in vitural set drop by showing that and by showing that friend as a principal can qualitate a quantitatively predict the self-organization of subsequent uh plasticity uh only using the initial data and as the modeling we can extend uh those canonical Network modeling uh to action generation another planning via the delayed modulation of heavy and plasticity and finally we discuss a possibility to extend this canonical model to Modern the social or shared intelligence to to interact with Mars per Partners so that's all of my talks thank you for listening uh this is

acknowledgment our collaborator and findings and our unit are now recruiting uh researchers so if you're interested in free check awesome thank you for the presentation to kuya um I'll just ask a few quick questions from the live chat and a few other things that come up so Dave says takuya used the phrase after plasticity was established was his group able to modify EG increase plasticity or is he saying merely that it was shown that there is plasticity or yeah I'm not sure if I understand correctly your question but uh those groups showed that uh the group showed that dopamine adding dopamine after two seconds after the association occurred can change the the the the the the the magnitude of plasticity so without uh how to say uh if dopamine are this addition was before this Association then processivity level is low but uh after the dopamine Addition after Association can change this level can increase the plasticity like this or like this so which indicate the the post-hoc modulation tool I think that answers it what are you excited for or what are your hopes or or feelings on where the octave inference ecosystem is at and where we're headed in the coming months and years but so sorry uh again please it's uh just what are you excited about other than your own research directions what are you excited about in the active inference ecosystem yeah so of course One Direction is the the modeling of uh social interaction so which is much a rich architecture than uh the interaction between the static environment so if both Asian Quran with each other then uh many interesting phenomena can be observed so we are excited uh with modern uh Zords phenomena using barot Square plausible uh Network neural network model through this equivalence any last comments any other comments that you want to make thanks again for the presentation and people should check out live stream 51 where you and I talked a few other times and went into some of the details on that work there's a really there's a lot there it's really exciting all right thank you see you next time bye see you next time all right the next session is going to be with Shanna Dobson this section is going to be called dark Imaginarium shared intelligence as an Infinity curiosity type all message Shannon make sure that everything's good with the audio wow talk so far welcome China how are you doing Danielle doing good um awesome well please take it away thank you for you know if you see my screen okay Yep looks perfect thank you okay perfect hey everyone thank you so much for the invitation it's such a great honor to be here I'm learning so much I'll be talking about um an idea I had called Dark Imaginarium where I'm trying to make shared intelligence the concept of that what I call an Infinity curiosity type so you were so gracious to link the paper so anyone can find more details um there on fill archive so I want to start with an discussion about ontological commitment so my first question is might we re-examine our ontological

commitments so in particular two of them one um Sean Carroll said so amazingly what cannot be known because it does not exist so Sean Carroll says the Heisenberg uncertainty principle is often explained as saying that we cannot simultaneously know both the position and the velocity of any object but the reality is deeper than that it's not that we can't know position to momentum it's that they don't even exist at the same time the second one I want to examine is Lenny suskin's quote we are all behind someone else's Event Horizon so suskin has a brilliant idea of space-time emergence as the entanglement of two black holes and what emerges from that is called computational complexity the role is quantum so every Theory should be derivable strictly from the laws of quantum mechanics one is going to give us the notion of shared two is going to give us the notion of ecology so what would an ontological commitment to dark look like so my continual goal as everybody knows is to sustain a simultaneous experience and to understand why I currently cannot to sustain a simultaneous experience would be to bypass the integrative system recall Consciousness so how would you attain a two Consciousness in the sense of in category Theory or an end Consciousness how would we actually develop something called like a two memory so my colleague Robert brettner and I fleshed out something called an awareness where we're sort of examining this idea Mike Levin and all have repeatedly shown that planaria memory is not in the head where is the memory is memory a sustained superpositional of what and also what's the structure of a nemesis could you have a superposition between memory and Analysis so I had a small dialogue with Chris Fields the amazing Chris fields and I said you know what is the difference between these two questions what is the waking State and what does Consciousness or could it be that these are entirely separate questions so we're sort of in a mess either way if there is a difference in these questions then it is possible to sustain a waking State without consciousness and or a conscious state in a non-waking so imagine somehow being in a waking State under general anesthesia well if these are not entirely different questions and we could contend that Consciousness is a form of waking state or something along the lines of a waking State as a condition of a conscious state so how can humans have deltaways which are canonically indicative of unconscious deep sleep and thus a non-waking state in the REM State the patterns of which are indistinguishable from a waking state so we all know that general anesthesia something like a propofol can reduce the polyrhythmic brain activity to one uniform hum and we know there's as many theories about what anesthesia does to Consciousness as there are anesthesiologists but we've repeatedly positive Consciousness as the produce State and I wanted to turn this idea on its head like is consciousness so robust or is

it a delicate interference or is it a delicate inference no we know Beginnings are always tricky but what if sleep was the proto-state so if you know if life is oceanic and it revolves from these deep pressures with the pressure of of getting onto land like squeeze the REM State into this full-blown wicking state that we're engaging in right now anybody who's listening so the waking state would be something like an acceptance which is I think quite impressive so it's in this mess that I attempt to Define dark Consciousness as a mixed frequency state of being self-aware or conscious while simultaneously being in a deep sleep specifically in three sleep so the sleepless protostate is quite is quite strange the REM brain activity is indistinguishable from waking activity and this sort of Enchanted it's Enchanted me for a bit the only difference is eye movements so either blinking or versus lateral movement so REM is naturally having mixed frequency brainwave States but the nrm states and the REM states are starting to blur as delta waves are creeping into the REM state now they've also found that sleep is a local phenomenon so of course I'm going to ask the Shanna questions so then who is sleeping who is remembering who is dreaming lucid dreaming is a Quantum superposition or is it a superposition or an integration so in a sleep State you know can then can we sustain a simultaneous experience so must be I be knocked out in order to sustain this level of simultaneity dreaming we know this characteristic of unsynchronized brain activity so is the question then my eye cannot just seen a multi-consciousness but my dark eye my sleep I can you can reframe this in terms of a reporting mechanisms and say perhaps we are conscious of things we don't think we are conscious of so let's talk about what is dark um a canonical definition of a dark Theory or anything like this is like um you know Dark Matter does not interact electromagnetically with matter but I want to sort of play around with another idea of dark um coming from De loses logic of sense something he calls the Paradox of infinite becoming and which I'm going to read on the next slide so what if dark was the simultaneity of a becoming that eludes the present let's hold on to that for a second is darker resolution issue but resolution is merely translation breaking translational symmetry so is dark a symmetry breaking issue and if so what symmetry and are we actually looking at the wrong symmetry is dark a sustained Quantum superposition that of what again is dark of many worlds Branch as dark the dolphin uni uni hemispheric sleep mechanism so I'll read from De loses logic of sense Allison through the Looking Glass involve a category of very special things events pure events when I say Alice becomes larger I mean that she becomes larger than she was by the same token however she becomes smaller than she is now certainly she is not bigger and smaller at the same time she is larger now she was smaller before but it is at the

same moment that one becomes larger than one was and smaller than one becomes this is the simultaneity of a becoming whose characteristic is to elude the present and so far as it eludes the present becoming does not tolerate the separation or the distinction of before after or of past and future it pertains to the essence of becoming to move and pull in both directions at once Alice does not grow without shrinking and vice versa so good sense affirms it in all things there is this determinable sensor Direction a paradox is the affirmation of both senses or directions at the same time I continue the Paradox of this pure becoming with its capacity to elude the present is called the Paradox of infinite identity the infinite identity of both directions or Senses at the same time a future in past of the day before and the day after of more and less of too much and not enough of active and passive and of cause and effect it is language which fixes the limit the moment for example at which the excess begins but it is language as well which transcends the limits and restores instant to the infinite equivalence of an unlimited becoming hence the reversals which constitute Alice's Adventures the reversal becoming larger and becoming smaller which way which way asks Alice sensing that it is always in both directions at the same time so that for one she stays the same through an optical illusion the reverse all of the day before and the day after the present always being eluded Jam tomorrow and jam yesterday but never Jam today the reversal of more and less five nights are five times hotter than a single one but they must be five times as cold for the same reason the reversal of active and passive do cats eat bats is as good as do bats eat cats the reversal will cause and effect to be punished before having committed a fault to cry before having pricked oneself to serve Looking Glass cake before having divided up the servings all these reversals as they appear an infinite identity of one consequence a contesting analysis personal identity and the loss of a proper name the loss of the proper name is Adventure which is repeated throughout Alice's Adventures hints infinite identity so I take this and I'm going to formally Define dark Consciousness as a state of being conscious while simultaneously being in deep Sleep specifically in three where dark is going to be a two-fold hybrid the simultaneity of infinite identity and of a becoming that eludes the present so let's reconstruct the dark Consciousness version of De los Paradox of pure becoming for Alice in dark Wonderland and through the dark Looking Glass okay when I say Alice becomes sleepy I mean that she becomes sleepier than she was by the same token however she becomes more self-aware than she is now certainly she's not asleep and self-aware at the same time she's asleep she was self-aware before but it is at the same moment that one becomes sleepier than one was a more self-aware so dark Consciousness I contend would take place in something called like a dark time

this is going to get weird so uh Chris fields and I talked about entropic time where a time where learning exceeds forgetting so we all know that linear time is based on the linear ordering property of the positive integers okay so what would be the structure of a time that does not interact electromagnetically with this waking State interface so like AI must pay attention all the time would this sort of be like a dark time sleep is local time is local we know what how beautiful a Time crystal is so time Crystal you know has a structure that is periodic in time can you play with that and say time is periodic and what so if we've explained that dark is not concerned with these Notions of before and after then we would want a notion of time that can support the Paradox of infinite identity with a notion of duration that is one of self-similarity so we can tend that dark Consciousness would occur in something like a periodic time which was a time based on the P attic number system periodic time would measure periodic duration okay it's been on all right so there exists few objects more Looking Glass than a piatic clock well maybe piatic go and collabi out chess and perfectoid go but after all the periodic numbers have no notion of linear ordering their shape resembles a sierpinski triangle which is a self-similar set hmm so this periodic time is not concerned with Notions of before or after that is a periodic time does not tolerate the separation or the distinction of before or after of past or future so because the Pediatric topology is one of total disconnectedness periodic time is more like a time of now continual zooming in to more and more now thus periodic time measures periodic duration through its fractal properties of zooming in so let's try to look at a piano clock all right so Pi time can take two forms the canonical periodic time defined on the previous slide and also topological periodic time which builds into its very structure the piatic topology of periodic time so there would only be one unit of time in periodic time we should call the Archimedes this is the periodic valuation so you can think of hour equals periodic valuation equals the Archimedes how strange it is to have only one unit of time topological periodic time measures duration topologically as total disconnectedness so the unit of time on topological periodic time is what I call the pitopo so we State this formula there's only one unit of time in topological periodic time called the pitopo so a topological hour equals a periodic topology called a p Topo so given such a periodic clock let us imagine the seasons in periodic time Panic snow Panic ring mean piatic weather types and piatic rainbows daylight savings time could take the form of changing the p in the piatic this could be very drastic but imagine pediatric metabolics periodic ATP and pediatric DNA imagine periodic cognition types like piatic memory piatic font piatic attention learning and pediatric perception imagine a pediatric Looking Glass imagine through the

periodic Looking Glass so it's pivot to fep what does ivp actually specifically say about Dark Matter even dark in the way that I'm using the term is there a free Dark Energy principle how can we strictly derive sleep from quantum mechanics according to fep does dark energy actually exhibit entanglement how do we test this we should be able to derive the entire active inference formalism straight from quantum mechanics alone uh by way of entanglement entropy some form of quantum gravity can we do some sort of dark General adversarial Network um to look at something like a dark brain we all know that lower lateralization in the brain is often associated with schizophrenia or you know people like Einstein or someone with a large parietal lobe probably like myself any mathematician and what Universal lateralization look like I think dark neurons would take the form of something like you treat the neuronal networks as algebraic curves or isogeny graphs you talk of neurons in terms of their properties as varieties so you're actually bringing the heavy machinery of algebraic geometry into neurology and my big goal is to actually create something like a neuronal Time Crystal so that shared intelligence would actually be a periodicity in time I'm not there yet so until I get that and it's like well let's just let's try to model these wild mixed frequency N3 plus self-aware states of dark Consciousness dark mathematics is having a coin it's gonna it's like a biomimicry which mimics dark energy does not you have to find the mathematical equivalent of that so I think you can do something like a dark dilethyism or just make a model that is modeling the mixed frequency states which is what I'm trying to which is what I tackle on the first in the paper so we've previously shown that Intrepid categorizations are condensed sets condensed sets form a topos so in the paper I outlined the construction of three potential mathematical models of this means of these mixed frequency N3 sulfur States um you can do dark Consciousness as a dark two pose which is a graph indeed two posts there's a two category of two sheaves over two site elucidate just a little bit dark Consciousness is a perfectoid length space in the sense of shows or as a diamond-like space Also in the sense of Schultz so I contend that the two sheath structure can actually model the mixed frequency state of dark Consciousness the N3 self-aware State and the Sheaf is a one stack again which is a sheath that takes values in groupoids not sets so this is already radically different and consider a two category of two sheaves so two category consists of objects you have the one morphisms between the objects and you have the two morphisms between the one morphisms you can keep going three category up to n category so this two category of two sheaves would have two sheets as the objects one more fisms between the two sheaves two morphisms between the one morphisms and so using that structure I think that you can actually model one

morphisms as what I'm going to call dark reflexivities uh these are going to be your local coherency States and then two morphisms are actually two inferences which are going to be Global coherency States amongst the mixed frequency States so a lovely property called Morita equivalent science states that inequivalent sites have equivalent Chief toposes and I do believe that that's the main property you need to get this local Global Behavior around neural networks so if you actually wish to model the mixed frequency clusters as fractals exhibiting some kind of self-similar behavior then you can use this Rich structure perfectoid spaces and are there sparkling successor the diamond so just a quick definition let perf that'd be the subcategory perfectoid space is a characteristic p a diamond is a pro étale sheaf on the site perf written as this quotient of a perfectoid space X by a pro-étale equivalence relation so a perfectoid space is going to be this étale space in the sense of Huber that's covered by étale spaces of the form $\text{Spa } R$ plus where R is a perfectoid ring and then points of $\text{Spa } R$ which is the étale Spectrum R Plus or equivalence classes of continuous valuations on R um summer National Sly diamond and you give incredible um explanation that was a parallel to a mineralogical diamond so you can let C be an algebraically closed étale field geometric points $\text{Spa } C$ to D is made visible by pullback along a quasi-pro-étale cover that results in profitably many copies of $\text{Spa } C$ so you're making a geometric Point visible as profitably many copies that was invisible before does the following parallel with a mineralogical diamond so the interior points are made visible as impurities which Sparkle as colorful Reflections on the many sides of the diamond um you can also do something maybe like a dark diamond redesignated functor is dark as a means to actually let's go the other way and see the impurities so then you're going to take the geometric Point make it visible Maybe by a pushed forward along with positive prototype cover resulting in periodically many copies of small C right so um a diamond is a they're using a diamond as a Sheaf but a sheaf in terms of a functor of points so the diamond spot QP is a functor from the category perv to the category sets you literally fix the perfectoid space and look at the home sets of X into the diamond and so you know the technically the scheme is then said to represent the functor so I want to Pivot to in Booking AI in this thing so my colleague and I Julian scape are coming up with um something I call I've coined the qubit pedagogy which is going to be a new model pedagogy that is actually infused with heavy principles of quantum mechanics and so that with these um we've we've created these Quantum intrinsic curiosity algorithms that are feeding into the aquibit pedagogy and based on those the Curiosity algorithm algorithms I've designed three dark versions so what I call the dark planaria curiosity algorithm it's a curiosity type it enables

encourages the AI to explore patterns of a planaria regeneration the AI would develop Quantum air correcting codes which mimic planaria regeneration so dark one the second one would be dark Quantum reality monitoring Network this is a curiosity type it'll courage encourage AI to explore dark reflexivity is what I call fractal identity AI would create models of n reflexivity in the sense of n category that's a new type of reality monitoring of fractal identity in a dark contrast state the last one would be some sort of dark Quantum N3 and REM curiosity algorithm this curiosity type is going to encourage AI to explore superpositions of various brain patterns in rem states to predict new types of N3 delta waves that could emerge in rem States so imagine AI could then develop new stages of REM sleep such as like one REM two REM of two and REM which is a very which is which is a clever limit and play on nram anyway the AI could also develop new dark cognition types that correlate with the new types of delta waves substitutional fractal identity seems odd but I do contend that objects living in fractal time the correspondingly have fractal identity it's a concept of outlined as follows so fractal identity has a reflexivity relation that is fractal so the very canonical loop the loop relation of an agent reporting back to itself I report back to me would be fractal and how strange is that so fractal identity would not suffer the same problems with continuity over time um since the time in which it exists has no linear ordering it would definitely suffer different problems but continuity over time is something Chris fields and I have been like how do you how do you explain that um so you can best model the fractal identity properties as perfectoid-like or diamond-like there and construct a notion of reflexivity as a perfectoid space the spaces are extremely rich structure but like kind of difficult but if you want to model fractal identity I think you should use them so this we create new concepts of perfectoid reflexivity diamond reflexivity so in in reflexivity the reflexivity relation could now be an in stack thus in reflexivity is an in-stack perfectoid spaces so here comes the culmination of this talk which is about shared intelligence is what I've coined the dark Imaginarium so imagine an AI that could C simultaneously compute the infinity category of every concept you know having some Quantum computation around to help the computational complexity what um you know uh not be that so dark Imaginarium is what is going to be what I call um an Infinity curiosity type so Infinity curiosity sites this is a higher order curiosity AI algorithm that encourages the AI to think in Infinity categories so that is it actually encourages the AI to construct an Infinity category as its means of higher order inference so objects are higher dimensional data sets the AI the AI would seek out higher order n morphisms between the objects and then morphisms between the morphisms so such dark Imaginarium is a meta curiosity algorithm that can create

its own Infinity curiosity algorithms and they're in creating curiosity types of the complexity that super fascinating most likely unknown to humans so it's functorial um what can what can dark Imaginarium do well the AI would develop inversions of current brain waves in their hybrid combinations it could develop in Delta wave in Theta wave and Alpha wave and beta wave and in gamma wave for n equals zero one all the way up like a two gamma wave that's structurally resembled a two category a three beta wave that's structurally resembled three category an AI could develop new types of waking States and new types of sleep States based on the combinations of these beyond the standard in one and two and three in rem and one REM AI could develop an end waking State an nrem State uh you know a three REM stake consists of three and one three and two three and three imagine an end dolphin I was capable of in hemispheric sleep State and other exotic life forms right what else could it do well it could develop new geometries for the brain patterns and for currently existing brain patterns typical shapes include sinusoidal non-sinusoidal saw² spindle K complex but maybe this AI could develop 3D and 4D versions of the canonical wave patterns like a tesseract gamma wave so but also the AI could develop entirely new brainwave geometries that correspond complex surfaces like a riemann's surface colavia Yao manifold or algebraic variety so there I could build new cognition types other than the standard five it could produce incognition types such as infant inattention in perception in learning in memory so it could also predict non-human species cognition types such as octopus thought or dragonfly attention lastly you can produce hybrid combinations of cross species cognition types such as like Optimum memory or Dragon man perception and then it would encourage the AI to develop new senses so you could develop one stack versions of human senses um like two seeing two hearing two tasting two touching two smelling each of these two senses once again take values in groupoids and not sets so you could develop instances of non-human senses as well and hybrid combinations like two shark human smell so since the aics and morphisms between the various and minus one morphisms the AI can seek to develop new language models based on higher order inference types so from these inference models the AI will explore Infinity languages and develop extensive prototypes you could use advanced mathematics to create new models of time that could support the new language models like two topos time Diamond time and then the AI would seek to develop new fractal identity types from the language models and it could design Advanced fractal Prosthetics such as a two-ear or a $2i$ to extend cognition so let's go a little bit further about this extending cognition these are just a few examples of what dark Imaginarium could conceive of so we could continue positive perfectoid version of this using

these new cognition types mental States and brain waves new geometries we can construct new patterns of thinking which would support neurodiversity in thinking so as said shared intelligence this is how I'm seeing it would be re-examined is this Infinity curiosity type you can go even further and construct a new concept of in shared intelligence so you'd have a zero shared intelligence a one shared intelligence and as well as the New piatic Concept of like P shared intelligence for p equals two three four using piatics let's go further right we repeatedly contend that one way to help Advance the way we think to extend human cognition is to upgrade the number systems upon which the canonical concepts are built upgrade them to piadic perfectoid diamond-like versions unless we get me thinking so imagine like a Braven wave pattern that resembles a collabio manifold could we actually reverse engineer the levels of conscious activity that could create such a rhythm how about mixed frequency brainwave patterns that resembled a perfectoid diamond what about an an in DNA molecule shape like a collabio enfold or an in ATP molecule or an in exciton condensate structured like a collabigo and fold so clearly different geometries of the fundamental molecules and processes of life would give rise to vastly different fundamental units of life and metacognition types the question is can we use dark Imaginarium to reverse engineer the complex life forms of rising from such extraordinary structures and the more and more complex cognition patterns what about a general adversarial Network that's tweaked and has Infinity categories you have two Infinity categories talking to each other developing their own language now Gan is adversarial using a generator and a discriminator with two neural networks contesting by a zero-sum game what about a collaborative Network so can we construct a gcn containing two diamond-like Infinity curiosity types and paradiamond with a dark diamond with a Quantum curiosity algorithm and its reinforcement learning so shared intelligence is this Infinity curiosity type shared intelligence GCM could create Infinity languages that are superpositions of the diamond dark diamond and be getting new models of intelligence you know higher order simultaneous computations by Quantum curiosity would go by Maybe by homotopia instead of as cost functions so it's futurecast just for a second um what is the end of a bill and the what is the inevitability of dark ecosystems mathematics of shared intelligence what can a piatic Infinity curiosity type prophecy is every perfectoid space Diamond two topos encoding the mixed frequency patterns of an incognition type in future work you may develop dark mathematics using perfectoid diamonds as dark diolethism aligning with a structurally dark with what is structurally so while we currently cannot perfectly model dark Consciousness perhaps one day one dark day mediated by the dark Imaginarium we will have the ethos and

the quantum computational complexity to do so thank you Shawna amazing presentation I'll just quickly read some comments and see if you have any thoughts from from the great comments so Scott David says I love the presentation lots of fodder for rumination thank you for inviting these Concepts into an intriguing choreography perhaps dark energy is our perception of the fact of information increase under free energy principle processes and since the universe is made of information dot dot dot Scott thank you so much yeah I think you should um yeah I think dot dot dot is um is a beckoning that you should or we should work on that idea I absolutely agree I'm glad you're inspired with these Concepts um I don't really know where they go but hopefully it's portals to something like that great effort one one more Scott David's question does fractal identity reveal the recursivity of rhetoric language Consciousness informing individual identity from Community identity inputs yeah that's wonderful so yeah my um my own thought is like what is an individual identity right Scott so the the reflexivity curve is always sort of stumped me how there's so much distance between I and me um and so it assumes this notion of continuity over time and so we've I've always said what sort of notion of time are we talking about to even have a consistent identity somehow you always wake up and everything is still here somehow you wake up and you're not a lobster I don't really understand how that works you know but um yeah that's nice if if you could use fractal to get something singular from something communal that might be cool but maybe fractal is something sort of like you're both at the same time and that's what I don't know yet so I think it's a totally revolutionary thinking to not have a before or after so if you can construct identity without before or after then that's sort of what I'm trying to do but yeah that's cool if you want to get identity from communal I think you can do that they may also sort of um Allure well this is truly food for thought and uh I really hope people can live with the darkness take it and run with it because there's so many fun ways to go thank you so much I hope so too so you're always welcome back thank you and talk to you later much bye good night bands onto the last session of the first interval greetings Nikki hello good morning yes good morning how goes it yeah I'm doing fine thanks um would you like to present it then we'll discuss uh yes that sounds like a good idea um let's see just going to share my screen yes can you see the presentation uh not yet okay let's see what's next Yep looks perfect okay great yeah should I do stocks yeah go for as long as you want and then anyone can ask questions in the live chat and I'll write down some things too thank you um yeah yeah so my uh presentation today will be will be on advances in machine theory of mind and a theory of my sophistication I'm currently a student at the University of Amsterdam and finishing my

Master's pieces on this topic and I'll be giving a broad overview of the machine theory of mind, an idea or concept for an active inference approach based on a previous work related to active inference theory of mind, so to start okay sorry. Okay, so for a global overview we'll start with giving a short basic definition of what theory of mind is and how it's been used, then I'll give you a global overview of current approaches to designing machine Theory of mind, including various approaches using biophysical models, etc., then I'll zoom into active inference models of theory of mind and social intelligence and collective intelligence more broadly, and I'll get into the issue of fear of my sophistication, which is basically the depth of recursivity of beliefs that you can have about another person's mental states and their thoughts or beliefs or goals about, and then we'll get into the issue of how to implement recursive beliefs and as an example we'll discuss Bayesian theory of mind with K levels of depth and then we'll get into the more specific Paradigm that I'm using right now to build a state of mind model which is based on a very simple model of the matching process Paradigm, so we'll talk a little bit about that and then we'll get into the model itself and current ideas for implementing the theory of my sophistication using active inference, so first of all what is theory of mind, fear of mind has been introduced by the very famous paper by Premack and Porteus in 1978. I think it was called "Does the chimpanzee have a theory of mind?" and they define fear of Mind as the ability to impute mental states to himself and others as a system of inferences and they say system of inferences of these clients probably do you properly viewed as a theory first because such states are not directly observable because and second because the system can be used to make predictions specifically about the behavior of other organisms, and I think there's already a lot in this definition and a lot of authors looking at their mind right now still use the same view of theory of Mind as a theory, a structured Theory of different kinds of mental states such as goals, beliefs, and desires that facilitates the prediction of the observable behavior in other agents such as spatial expressions, movement, or speech [Music] and that's yeah I find it very, how much this aligns with, maybe a predictive processing view of social perception, but also how weird it is that this Theory of theory of mind is still so widespread in the theory of my literature modeling literature so there's been some discussion on whether you can properly regard serious mind as a theory and if so what kind of theory it is. So within the philosophy of mind they've been basically two mainstreams or main perspectives and theory of Minds according to the plastic Theory theory of Tom, fear of mine is a cause of fear of how these internal mental states generate

other agents behaviors so it's the same kind of view that we just saw um and then the opposite side is called simulation theory of theory of mind which posits that we use the same kind of cognitive and neuronal resources for understanding and predicting other agents as though those that we use for understanding the cell and simulation Theory can also take on different forms and just the sun is coming up okay nice to them so um that's why I'm breathing light at the moment um but basically a lot of versions of this some process that we use the same kind of action prediction mechanisms for ourselves for others um and there is this new version of simulation Theory and that Gordon proposes in 2021 which very broadly says that the um we have the same kind of theory for other agencies for ourselves so there's this generous theory of what it means to be an agent and that is refined at different stages to facilitate complex sensitive predictions in social interactions so this is also interesting to look at when you model theory of mind what kind of a um what two things theory of mind is and how the specifically implementing um so why would we be interested in creating computational models of theory of mind and so I think there are three um important reasons why this is very relevant at the moment first of all there is a technical reason or the reason of technological advancement um there's been some proposals that um argue for for the development of artificial social intelligence or waiting the technical reason for for developing computational models of theory of mind is to improve artificial social intelligence especially as when we start to live in a world in which research on human AI interaction becomes more relevant um and that's been in this paper by Williams for example that's argued that improving artificial social intelligence is needed for successful AI human cooperation and it could improve communication trust and maybe also value alignment um by utilizing theory of mind for Value language for example and there is a very strong foundational reason to study social conditioning generally um and the evolution of social cognition um for in silica hypothesis testing and clarifying the conceptual boundaries of um one great example for this is research by divanadol which use a computational model of recursive theory of Minds um compared to alternate models to study a different cognitive strategies in non-human primary primates and they find no evidence for the hypothesis that theory of Mind sophistication involved with an increase in her size but they do find a correlation with for example neocortex ratio um so in this way yeah their theory of Mind models can give us some insight into the evolution of social cohesion um on the clinical side um theory of Mind models can be used to understand relevant social variations in clinical calculations for example through computational phenotyping and Diagnostics and by studying social behavior in groups that have some difficulty um with social interactions um and Crews

that shows some kind of variation a social cognition such as people with Autism Spectrum people in the autistic Spectrum people with ADHD or schizophrenia so I hope that with some of these reasons I am convinced you that moving theory of mind is very um relevant at the moment and there have been a lot of people already on this um and I've created this um very Global overview of some approaches that are that are representative or some models that are representative of their more General approaches there's this um ringtone model which aims to basically build up fear of Mind abilities and uh abilities facilitate his social interaction from the bottom up using biophysical models of neurons firings um that's why I've placed it on the dynamic scale so I've aligned try to align these modeling approaches along the axis of have a trip time um how broad the description is from the individual level to maybe describing what broader social and cultural processes and along the level of description from the molecular eye physical to the behavioral um and one thing you can see is that there is a lot of work based on Invasion inference and active inference that moves along that has relevant ties to but to the molecular and neurobiological level but moves along the lines of making behavioral and cognitive predictions um and there is very there are very little models that really um also formalize the concrete implementation of theory of mimes um and there is also I think this need for connecting um this dynamic implementational level two individual level social permission and putting that in the context of social interaction more generally so some of the representation representational approaches that I've selected are the tall Nets which is which basically uses a neural network embeddings to train reinforcement learning agents we have a k Tom which is evasion recursive model which I'll come back to later um some or theory of mind with self other modeling uses multi-agent reinforcement learning with Goal inference and we have the more theoretical active inference model of implicit cultural learning course thinking through a device which could be seen as an alternative or a redefinition of theory of mind so to sum up um we have this stream of biophysical models um most notably brain tone which was developed by Xiao at all um and his strength are its neurobiological plausibility and making Dynamic predictions um about neural oxidation and in that's been um populations that are associated with uh theory of Mind abilities and its weaknesses is that it's quite computational quality and it it's probably difficult to implement in an interactional setting this there's some deep learning models such as tongnets by rabinovitz and serve other modeling which I just mentioned um that are also quite neurobiological and also make some Dynamic predictions um but there is this strong danger of shortcuts which has been addressed in deep learning models um that the tasks might not be solved by learning the relevance ability of

being able to infer another Asian schools or beliefs um but just by learning a simple lower level associations between for example the perspective or the um this how another agent stands and where for example a single reward would be so there are a lot of ways to solve the task and then assist finding that deep learning models sometimes these these kinds of shortcuts um so that's something that's needed to be addressed um then there are some Beijing belief-based models such as Ketone um for for recursive theory of mind um and the Asian Tom which was originally developed by Baker at all in 2011. um and its strengths are that it's quite efficient it's very easy to interpret and its weaknesses are that it that could brought to the implementational level is longer and it requires many priorities um and there's this question about whether it's stateable because of the pro specifications that you have to give um okay so as one of the I think the interesting newer developments on a theory of mind uh it's is a hierarchical active inference for collaborative agents which has been recently published and taika is based on variation theory of Mind by Baker at all but it adds this formalization of the leaf resonance which is the incorporation of beliefs about another's intentions with top-down um and this influence of these other agents beliefs are modulated by susceptibility parameter which controls this influence um and what I find so I think it's important thing out here that they used an approach based on asset inference but it's not the they don't use the spare energy principle for example um so they do couple perception and action but just in a different way using a common filter so the way that formalizes is slightly different but I think the overall structure can still be um yeah informing or inspiring for active inference and Essence so what is interesting about their um simulation results is that they show that agents um that there's some emerging collaboration on tasks for local use of this susceptibility parameter without any explicit planning or communication needed and this means that they improve upon the original Bayesian model which explored um different sort of overall plans for all of the agent which means that an agent employing the Asian Top would need to reason um about the rules of all of the other agents and which was very costly and they were and the authors wanted to improve on that team model maybe on the Fly coordination which is more spontaneous and takes less time um and they also show that incorporating the least resonance has a very positive effect in scenarios in which there's unbalanced information they did this examination the overcooked domain in which agents have to have agents have to collaborate on different cooking class and the goal is basically to order the cooking order um and then the intention would be any kind of action that has to do with completing the order um and they show that incorporating the leaf resonance um is especially it seems to be especially useful in situations

with antennas information which for example One agent knows the order and the other doesn't um and these agents develop some kind of leader follower Dynamic um in which one is most sensible to the other agent's intentions and one agent attempts more to the evidence um from from the environment and this also aligns with what we know about here it might mainly being employed in situations in which shares some asymmetry in what the agents know about the environment so I thought it was very um insightful okay so I'm going to um go into uh some active influence approaches that I think are relevant for developing activated Spirit mind uh first of all there's this very real proposal on inculturation and normal developments thinking for other Minds a variational approach to cognition and culture which we'll go into a little bit uh we'll go into a castle and past paper on ideas for spreading a free energy proposal for Community cultural Dynamics um will go into Kaufman adults model on prolexive intelligence and finally also the uh proposal by muchuses at all on Beijing inferences about the self and okay so first of all I am starting with based inference about the cell and others um because it gives the very sort of full introduction to I think predictive social cognition and that's also why I've added this figure on the right um that is from the time in Fulton Bank in 2018 which explores um the social the predictive social brain um and what kind of features um seem to be seem to be relevant for categorizing and understanding of its behaviors some figure on the right you see that there's a space this trade space which is Odin by power such as sociality and balance um and that constraints the state space which seems to be ordered by a rationality social impact imbalance and courses are just correlations but these future seems to be the most predictive of what they're seeing in the brain um and you can see that there is this sort of yeah uh basically it's a very strong top-down constraints of the possibility space in which you interpret other people's and you can also see that based on another agent's actions you learn something about where they where they fall uh within the state space in a given context and also where they form the trade space what kind of so um at the same time you might have already have some priorities on what kind of a person this is um but you're also learning about it uh you might also have some careers on how someone behaves when they're not for example or when they're sociable versus unsociable and all these kinds of cry is informative prayers they inform your predictions in the context and they make this task and much more manageable of predicting other people's behaviors on the Fly um and I still think that links very well and um that team which is this proposal which is that which is basically that inference is about the self and others at different time skills facilities interpersonal minimization of surprise um and thereby optimal social decision making so these uh and these

self and other representations first um that are the facilitates um personal minimization of surprise um could be seen as representations that inform uh our our inferences and predictions on the thigh I think yeah I think it's also important to mention that one of the main addition points of their paper is that um that self-representation so beliefs about how beliefs about the cells self um can uh facilitating personal minimization of surprise so even what I think about me and how I represent that while communicating will facilitate the other person's uh optimal decision making at that moment um maybe life is by providing the right kind of information that they need to Ultimate um okay so to the next uh two interesting papers obvious we're spreading um and an active infrast model of collective intelligence so the key idea I think for for both of these papers is the emergence of career behaviors from local interactions for customers this is the global uh group behaviors cultural truth and transmission or a transmission of beliefs and homes and they describe this process as the mutual achievement of actively inferring agents through communication and theory of mind is one of the many mechanisms that facilitate this process of achievement so that's the link and they showed that the spread of certain kinds of beliefs through a group ending models just by making agents interact in um in dyadic groups um and stiminally uh Craftsman adults show that the system you can create system of free energy minimizing agents with some minimal uh properties of social intelligence such as the ability to assess how it's self-similar another region is to themselves and the ability to inspire another agents intentions um and the ability to align their goals to another agent more or less um and they show that if you put all of these agents together in those diets they don't only minimize your energy on the dietetic field but they also minimize free energy as a collective and thereby this could this is a very simple model of how adaptive behaviors adaptive group behaviors could emerge from Agents with very simple social abilities so um now we get to thinking throughout the Mayans um which is not technically um a proposal for fear of mine but it's the proposal for how social cognition shapes inculturation more generally and what kind of abilities agents need to facilitate this kind of known formation and I think it's a very yeah interesting proposal because um I think it highlights an important function of social behavior more and you could even see it as a way to redefine theory of Mind as more of an interactional concept than as an individual level cognitive ability um so according to this proposal singing through other Minds is a property of multiple interactive energy minimalizing agents and a process by which agents infer other agents expectations and based on this the authors propose a definition of enculturation as a process of learning social expectations through the selective passening of attention that guides

agents towards relevant affordances within their social and cultural Niche um I'm not sure if you're all familiar with the concept of affordances but it was quite hyped up a couple of years ago in general it is defined as the relational construct denoting possibilities for Action um offered by an environment that are code determined and also by an agent's interests goals and abilities so I think the the standard example for this is when you are thirsty um a glass of water might appear as an affordance or you might not consciously or Michael's attend to it otherwise um and cultural sport instance um in this sense are epistemic affordances um so for instance that gives you some epistemic value that have come to stand out as reliable and relevant based on a history of cultural information they encode and good examples for this are different kinds of signs of traffic signs and we learn what they mean from a young age and we sort of inherited the symbolism of the church science and they guide our they guide our attention towards points in traffic and they regulate our behaviors so when you look at the complete proposal it sketches how the regulation of social behavior depends on the learning of shared laws and expectations um and also how this connects to utilizing cultural affordances um and thinking through other Minds as profits interactional property in which agents learn to infer each other's expectations to optimize Collective behavior and decision making okay so from these interesting active inference and broader proposals of to your mind to get to the issue of some sophistication in active inference so what is fear of my sophistication um the short Recaps it's the depth of recursivity that can be utilized for serious mind of the form I think that they think that I think and it has been found that agents generally use a representative of between zero and four um and um that humans humans ability to for recursive belief reasoning is generally limiting um but impressive performing this could go on forever um and one way to implement this kind of sort of uh arbitrarily recursive belief in fairness is composed by that at all um we use the citation model of sophisticated theory of mind um with cape home agents and provide stimulations in which Ketone agents take as agents um that are just one level of recursive recursivity under them and this model has been used for a lot of interesting uh simulation experiments for example um Divine at all showed that in a trade-based analysis so with agents have been fixed um fixed steps of fear of mind represivity there is a specific distribution of Agents with level one time a level two-tone and that could provide an evolutionary Advantage compared to populations with larger proportions of serotonin fatal agents so they use they look at being adaptive properties of systems with different kinds of ratios of theory of my sophistication um and in another in another experiments which is a state-based for sure it all showed that if you use computational signal typing or autistic

individuals um they show this shows reduced sensitivity to task framing um and strategy switching within the game that I've been using at the required sophistication and so these are just some interesting applications for fear of my sophistication and the promise of modeling theory of my sophistication would be that you could investigate how people flexibly switch between different social strategies while solving the sexual tasks and one of them would be to increase the depth of fear of line sophistication or decreases depending also on what you think about the other the other agents sophistication so what I've been working on at the moment is an active inference model of social prediction during the matching pennies game um the matching pennies game is a game farm economics um with a very simple format there are two rules you have the guesser and the Challenger the Challenger hides a penny in one of the two hands the guest needs to guess which hand if the guesser is correct they win um I think I guess it was wrong being challenges um this game has has some logical equivalence um one version of the game makes both of the agent celex heads or tails so the Challenger select heads or tails and the getter to guess if it's heads or tails but both have the same formal structure psychological experiments of subjects performing this task have shown that there is a strong social framing effect in a matching penis game so participants will be paved and it's full differently depending on whether they're seeing that they're playing against a human agent versus um an AI for example additionally a comparison of Cape Town and other computational strategies um that were fitted on experimental data of subjects playing the game revealed that participants seem to reduce their mind when playing game so there's a strong indication that fear of Mind use does their role in the matching pennies scheme um and that social framing will um influence the social strategy that's being employed by the players so this is the very simple zero term architecture for the matching pennies game and I included beliefs about cell choices um as also kind of yeah monetary self-states beliefs about other agents choices on whether the Asian things the uh the opposing agents agent has hidden depending left or right um and then the agent also tracks reward since the game in the setting is device in a way in which push the agents both making observation simultaneously and then all of the observations are revealed um this setup made sense so just going to drink a little water um the outcome modalities of the patients that the agents have are its own choice observations um the choice observations of the opposite agent and the reward observations and the possibilities there the action space is very limited and the seeker can choose to seek left or right and The Hider which is the same as the Challenger before um has the possibility of hiding and electoral rights and the reward is uncontrollable since it depends on both agents

choices at the same time yeah so this is sort of a summary of the architecture that I just described and this is another visualization of this setup and um you can see here that if you look at the likelihoods um the reward mapping for the Seeker is just the opposite as for the island um and I think that this simple setup is uh very suited for uh studying feared Minds in these kinds of simple competitive uh uh interactions um if you wanted to um study the use of theory of my sophistication in the matching pennies game um using active inference uh I ran into a couple of challenges um and one of them was how to how to steer the problem of recurrence so how can we make a difference between the focal agents um and the other agents if we try to model the other agents within one agent um especially if you want to introduce or formalize depth of paper complexity it can't just make one agent stimulate the other agents stimulating their cell there needs to be some kind of sort of structure that distinguishes sort of the Soul's inferential mechanism and self-learning mechanisms from what you what the agent focal agent thinks the other agent is uh thinking and seeing and doing um and I think the gentleman with the bowler hat by first in 2003 is a good example um for this problem so you can um that the basic setting is that our world is populated by beings like ourselves um experience experiencing themselves as a subject as the focal point of their their worlds with their own thoughts and feelings and they all think that they're the one and only subject um but in but their cons will be so there needs to be sort of an either an outside focal point um or one ultimately subject um and um yeah so I think more practically the questions that arise from this example are how do we distinguish in focal agents and stimulated agents um and how can we simulate other agents in a way that is computationally efficient and maybe uses the same kinds of resources that the agent already has so I actually had a love meeting um at the theoretical neurobiology lab a while ago and we were brainstorming about this issue um and one of the ideas that came up was to use sophisticated Insurance um fourth year of mind so in this case um an active inference formulation of the leaf repositivity could be used for modeling recursive view of mind um so basically the planning using predictions about the other agents into your class accounts would consist of evaluating all the kinds of hypothetical scenarios all kinds of contextuality on how your actions like influence their beliefs and their actions and so on and so on so this would be one bro to go into but this would also mean that all these conceptuality will need to be evacuated um so I was thinking is there a sort of more efficient way to sort of use these kinds of either internal simulations or confessional evaluation um and the current uh approach I'm working on um which I've also sort of yeah which I'm also still developing so I'm very much open to feedback and um comments on how this could work

um and possible alternative approaches at this point um but the general idea would be to create a possible internal stimulation of the other agents control mechanism so taking the learning mechanism of an active influence agent as sort of the definition of what the focal agent is that that is the agent that perceives and learns and impairs from the um what the agent does when predicting the observations of the other agent or the predicting the other agent's behavior um it's basically performing partial simulation about what it thinks the other agent's preferences are and what it thinks that um the other the other agent's strategy um which is encoded by the transition from state to state and one of the potential ways in which she could include recursivity in this way is that you pretend that you have control of the other agent's actions and you predict the other states you predict self-states um it evaluates um yeah you evaluate the state action probabilities um choose the most valuable action based on the distribution um simulates the other states that come forth and simulate the self-states that come forth um and then you have a new and then this can be used for the next round of action selection by the other model um so in this case only the prior preferences and the agent specific transition probabilities would be needed as sort of information on the other agents and because you also want to include some kind of learning about the other strategies one of the things you could Implement um learning also that they made literacies and so having sort of a specific uh the um having the agents accumulative evidence about this transitions over time to also better predict the other agent's potential actions so I asked sort of the basic principle would be to simulate control from the perspective of the other agents and the recursivity comes into play when you stimulate the other when you've already simulated the others predicted actions and predicted other states and you've simulated your own actions as a consequence of the stimulated actions you can feed that back into the other model to predict what they would do if um they thought that you were in the state that you were in um and I think if you sort of uh yeah I think that could be a yeah a way to develop recursivity um in a way that does not need complete simulations of parallel agents within the same system um and of course there should be some kind of temporal discounting so parameter that discounts evidence based on the simulated planning arising um that is sort of yeah that is that defines by the key number of steps um so yeah this is sort of a conceptual outline um of how you could Implement uh um recursive fear of Mind using active um we haven't been able to disclose the Nations using this yet but we're working on that and uh we're very um much open to discussions on this and Technical comments on how to improve on this outline um and one of sort of the the open questions is still um that you're still thinking about is how to make the agents flexibly infer the

depth of repressibility of the other agents um in the setting um yeah Special thanks to uh Yong Dumas who also be there at the panel I think in session two um we're super we're supervising me uh during this project uh thanks to the ppsp level I'm doing my internship to uh really excite who's also working on a related project uh the pi mdp team for that great great coding package that I'm using um and for call First and for his feedback and ideas on my previous presentations thank you very much for listening oh great presentation very um comprehensive review of a lot of the different theory of Minds options out there um well if anyone has questions they can write it in a live chat what do you want to explore or can I ask a question or or what what would be fun for you blinded by the lines at this point so I'm not very um I can't easily read from my screen so maybe it would be nice if you could um if you could read out the questions that are in the live chat sure oh yeah sure sure um well first one one part I found interesting you mentioned that people play differently when they are playing a human or when they believe that they're playing a human so what is that like what is the difference in play when we believe that we're playing another person so as so if I look at the original page where there is one study on this and then um they did the computational stenotyping study using Cattlemen alternative strategies um and they showed that if you use the social social framing condition that subject will subjects will more probably employ my strategy is compared to other simple simple reward based strategies so maybe those strategies that should be described using some kind of reward learning mechanism um but the difference in their overall sort of trajectory of choices is I don't know yeah a few few themes then that you brought up that came up earlier uh in the session so one was using pi mdp and sophisticated inference which Aspen Paul gave a walk through on and this question about recursivity and about the way in which you can on one hand talk about abstractly like k recursivity yet the biological system doesn't implement that kind of a logic and so so um could you maybe describe a little bit again how does the the structure that you laid out for the generative model capture some of those features of recursivity without falling into the Trap of um just theory of mind all the way down yes so my general idea was to simulate the other agent's action prediction um based on uh learning about the transition um probabilities for the other for our what we think the other um how the other agents stays transitioned um and then if you um you've predicted what the other Asians would do to do you can use that as a prior for calculating um action dependent uh expected States and based on those evaluating those expected States um we have your who stimulate your own choice mechanism and I think in principle you could use this recursive simulation um indefinitely but also yeah it's also device in a way that it uses the same kind of um the

same kind of sort of just the architecture of the local agent so the other idea was to just use internal simulations all the way down but that would yeah that would be very comfortably impossible and yeah I think it's a very good question because how why would you even need to formalize belief requestivity all the way down if in end effects um it will only use Tom with the with the sets sort of Maximum depth and the fact that we use theory of mind with a maximum death might also be sort of a hint that he might also use um a different or what we do might also be described by a different strategy so I yeah I'm not sure if that's an answer to your question yeah well when I saw this and some of the other generative models um you mentioned maybe here like you pretend like you can control the other person's and that's kind of the fundamental weaving of action into the inference Challenge and then that uses expected free energy well just like any other policy selection so it's like pragmatic value would be bringing that controllable State your your uh social Partners mind your beliefs bringing them into alignment with your preferences is pragmatic and then learning more about them is epistemic and so then that might enable some theory of mind like Behavior where it's like when you don't know enough to put the squeeze on then you have a kind of open-ended learning and curiosity and question driven theory of mind but then once one has enough to kind of exploit rather than explore then without necessarily even updating the model or engaging any like deeper recursivity or strategic shifting you could get um social behavior that's oriented towards control and pragmatic value rather than just like learning and questioning yeah I think you're you're very very the idea was also to enable the agent to um maybe first few strategies that are more useful in um Gathering evidence about the kinds of strategies that the other agent is using before exploiting um what you've learned about the other region so what are your next steps with the work um so I think yeah right now the main task would be to implement this using pi mdp and see if there are any sort of formal for climates that I need to make well in the implementation because it's yeah if you can't build the two cons you don't understand it so um that's what I'm yeah that's what I'm working on right now and then the next step would be to introduce um a mechanism to sort of maybe very uh recursivity um based on what the agent infers the kind of recursivity the other region has um but before that I also want to want to use the model to perform computational phenotyping subjects that were doing the matching honey sauce and had different kinds of diagnoses so there were neurotypical um Youth and um youth with um youth on the ocean spectrum and things with ADHD and anxiety and it would be interesting to see if there is any sort of difference in the depth of repressivity that they employ as sort of the flexibility with which

they switch between strategies that makes me think about different different levels or or differentials between self-knowledge and other knowledge on one hand we seem to know a lot about ourselves but also were often unaware or blind to some some features or regularities of ourselves that that might be highly attended to by others and so when we're rolling out these games or scenarios there might be some theory of Mind settings where like knowing yourself better carries the day but other settings where knowing like the game is what matters or knowing the partner in an iterated Game Theory or like just knowing having a lot of experience um like there's a lot of a lot of degrees of freedom in modeling these very um advanced cognitive ecosystems yeah I think that's also one of the reasons that we chose the matching pennies paradigm because the Paradigm itself is very simple what we're trying to model is quite complicated and on the matching pennies like um so it's in one hand you know the person has two hands behind the back and then they bring out the two fists and you're trying to guess like where it is yeah and then I wondered well what if it was you have to guess where it isn't on one hand that's like kind of the same or like if you say well you win when you pick it but then if you say well you win when you when you pick the empty hand mathematically it should be like the same because knowing where one is is like knowing where it isn't but but just as a human game player it has a very different feel and those kinds of like um reversals might give us insight into our priors that support not just open-ended abstract theory of mind but like kind of with a lean towards being pro-social or like heuristics in place that that support making um quick decisions together maybe rather than I I don't know yeah I think those are all very good very very sort of promising trains of thoughts I just I think in the case in which you you need to think the empty hand we might have very strong sort of beliefs about what if what or we might experience sort of be a presence of something as more inherently more rewarding than the absence of something because we have all these associations with gifts and getting things and like rewards are usually things that are there so I can imagine that in this case sort of the perceived you would you wouldn't perceive it as rewarding but formally it would be completely the same yeah yeah like uh receiving a benefit versus averting a loss yeah and then how that plays in and and and our encultured priors and our our personal histories and how these very deep layers of of differences might play out and uh what kinds of empirical data or online data could could help us generate like unique explanations or predictions maybe situations like you said some situations where a reward-based model might be sufficient but then I think the interesting question for active people is like where's the unique explanation or that that really um has active shine and shows where

surprise bounding as an imperative gives us like more realistic or more concise or conciliate explanations in a way that like it doesn't even make sense to appeal to reward yeah I think that's also the tricky bit about this Paradigm is that it's very I mean it's for war based in a sense but I think because you have this social framing effect you also have these interesting learning mechanisms um and mechanisms of uncertainty reduction that might play out in how the um the agents behave in the end but that's why we saw that it would still be interesting to use active inference to explore this Paradigm but on the other hand I think if you on the moment at the moment you're looking at individual variation um you might get interesting differences in verbal sensitivity or a version of yeah risk aversion stuff like that um that might be better expert in a different context oh any last comments or or thoughts no yeah thank you for uh for your comments and your uh uh insights yeah it's awesome it's a great Direction happy to see it thanks everybody for presenting in this first interval and see you in the second interval